

A power saving for square-difference-free sets

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Abstract

We prove that there are absolute constants $c > 0$ and $C < \infty$ such that every set $A \subseteq \{1, \dots, N\}$ with no nonzero square difference satisfies $|A| \leq CN^{1-c}$. This answers the fixed-power question posed by Green and Sawhney.

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1 Introduction

For a positive integer N , write $[N] = \{1, \dots, N\}$. A set $A \subseteq [N]$ is *square-difference-free* if

$$(A - A) \cap \{m^2 : m \in \mathbb{N}, m \geq 1\} = \emptyset.$$

Thus the restriction concerns differences in the integers. Define

$$s(N) = \max\{|A| : A \subseteq [N] \text{ is square-difference-free}\}.$$

Our main result is a fixed power saving in this extremal problem.

Theorem 1.1. *There are absolute constants $c > 0$ and $C < \infty$ such that, for every integer $N \geq 1$ and every square-difference-free set $A \subseteq [N]$,*

$$|A| \leq CN^{1-c}.$$

Equivalently, every subset of $[N]$ larger than CN^{1-c} contains distinct elements whose absolute difference is the square of a positive integer.

The exponent obtained here is positive and absolute, but extremely small. The argument does not yield a useful numerical value or an optimal exponent. All finite constructions in the proof are fixed independently of N and A ; their size affects the constants and the exponent.

Remark 1.2 (Square-difference colorings). Theorem 1.1 gives a polynomial bound for the classical square-difference coloring problem [2, Section 4]. Fix its constants c, C , reducing c so that $0 < c < 1$ if necessary. Whenever integers $r, N \geq 1$ satisfy $N > (Cr)^{1/c}$, every r -coloring of $[N]$ has same-colored elements $a, a + d^2$ with $d \geq 1$. Indeed, a largest color class has size at least $N/r > CN^{1-c}$. Only the two endpoints must share a color; no color restriction is imposed on d . Equivalently, if G_N is the graph on $[N]$ joining distinct integers whose absolute difference is a square, then

$$\chi(G_N) \geq \frac{N}{s(N)} \geq C^{-1}N^c,$$

since its independent sets are exactly the square-difference-free sets.

1.1 Previous work

The problem originates in a question of Lovász. Furstenberg [8, Theorem 1.2] and Sárközy [16] proved independently that $s(N) = o(N)$, using ergodic theory and the circle method, respectively. Sárközy's quantitative estimate has the form $s(N) \leq N/(\log N)^{1/3+o(1)}$; see also the historical account in [3]. Pintz, Steiger and Szemerédi [14, Theorem 1] strengthened this to

$$s(N) \ll \frac{N}{(\log N)^{c_{\text{PSS}} \log \log \log N}}$$

for an absolute $c_{\text{PSS}} > 0$. Their argument combines concentration on arithmetic progressions with an iteration that increases the collection of large Fourier coefficients. Bloom and Maynard [3] subsequently obtained

$$s(N) \ll \frac{N}{(\log N)^{c_{\text{BM}} \log \log \log N}}$$

for an absolute $c_{\text{BM}} > 0$, using bounds for the additive energy of rational numbers with small denominators to improve the density-increment argument.

Green and Sawhney [9, Theorem 1.1] prove

$$s(N) \ll N \exp(-c_1 \sqrt{\log N})$$

for an absolute $c_1 > 0$. Their arithmetic level- d inequality transfers product-space hypercontractive estimates to rational Fourier coefficients. Theorem 1.1 answers affirmatively the fixed-power question posed in [9, Section 1.1]. Adajar et al. [1, Theorem 1.1] extend the arithmetic level- d approach to fixed integer polynomials having a root modulo every positive integer. For degree at least two they obtain a density saving $\exp(-c(\log N)^\mu)$ for every fixed $0 < \mu < 1/2$, with constants depending on the polynomial and μ .

In the opposite direction, Ruzsa [15, Theorems 1 and 2] constructed square-difference-free sets of size $\gg N^{\gamma_0}$, where

$$\gamma_0 = \frac{1}{2} \left(1 + \frac{\log 7}{\log 65} \right) = 0.733077 \dots$$

His construction uses a set of residues with no square differences modulo a squarefree integer to restrict selected digits of an integer expansion. Beigel and Gasarch [2, Lemma 3.6 and the proof of Theorem 3.7] and Lewko [12, Theorem 3] obtained the improved exponent

$$\gamma_1 = \frac{1}{2} \left(1 + \frac{\log 12}{\log 205} \right) = 0.733411797 \dots$$

by exhibiting suitable sets of twelve residues modulo 205. Krachun's construction [11, Theorem 1] passes the three-quarter threshold and gives

$$\liminf_{N \rightarrow \infty} \frac{\log s(N)}{\log N} \geq 0.752796455874514 \dots$$

A substantial gap remains between this lower exponent and the upper exponent $1 - c$ in Theorem 1.1.

Reflection positivity yields multilinear inequalities through the chessboard method of Fröhlich and Lieb [7, Theorem 2.2] and its abstract formulation by Fröhlich, Israel, Lieb and Simon [6, Theorem 4.1]. Hatami [10, Theorem 2.8 of the arXiv version] characterizes graph seminorms through mixed Hölder inequalities. Conlon and Lee [4, Theorems 1.2 and 1.3] use finite reflection groups to obtain such inequalities for absolute-value inputs, and also for real-valued inputs when the two defining sets of simple reflections are disjoint. The signed mixed inequality is a classical feature of the chessboard method; its role here is to permit Fourier lifts that can change sign. We construct a finite-field probability law with the particular cycle support, reflection positivity and conditional mixing needed for the integer problem, and prove the signed inequality for that law directly.

The rational-frequency energy estimates of Bloom and Maynard [3] and the product-space methods of Green and Sawhney [9] are methodological precedents for the arithmetic part. The argument retains rational Fourier analysis and rescaling along square-step progressions. Here the quantity compared between scales is a multilinear functional of a signed Fourier lift.

1.2 Outline of the proof

The proof constructs a nonnegative quantity that dominates a fixed power of the density, and then proves that this quantity decreases by a power of the interval length. The comparison with shorter intervals is available for every residue specification. A separate estimate makes a fixed positive proportion of those specifications negligible; these two estimates produce the contraction.

Here is the quantity to be controlled. Choose a fixed even integer d and a fixed set V of d labels. For a finite set $\mathcal{P}$ of sufficiently large primes, put $X_{\mathcal{P}} = \prod_{p \in \mathcal{P}} \mathbb{F}_p$. We will construct a probability law for a tuple $(z_v)_{v \in V}$ in $X_{\mathcal{P}}$, and for a real function g on this space define

$$Z_{\mathcal{P}}(g) = \mathbb{E} \prod_{v \in V} g(z_v).$$

Its d -th root will be a seminorm satisfying $|\mathbb{E}g| \leq Z_{\mathcal{P}}(g)^{1/d}$. To apply it to $A \subseteq [N]$, take $\mathcal{P}$ to be the primes between a fixed threshold P and N^β , where $\beta > 0$ is fixed and small. Let G retain the rational Fourier coefficients of 1_A at frequencies of squarefree reduced denominator at most N^β whose prime factors all belong to $\mathcal{P}$, including the zero frequency. Section 5 defines this *Fourier lift* precisely. It is real and its mean is exactly $|A|/N$, although its values may have either sign. With these choices, write $Y(N, A) = Z_{\mathcal{P}}(G)$. The seminorm bound gives

$$(|A|/N)^d \leq Y(N, A).$$

Thus a fixed power decay for Y proves the theorem.

The tuple law is constructed separately over each prime field, then multiplied over the primes, collecting the field labels into vectors $z_v \in X_{\mathcal{P}}$. Every individual field label is uniform. The law has two further properties with distinct purposes. First, it is reflection positive: specified involutions of V exchange two halves, and the matrix coupling the half-tuples is symmetric positive semidefinite. Repeated Cauchy–Schwarz across these cuts proves the signed mixed inequality and seminorm bound (Proposition 4.1). Second, outside a part of mass at most p^{-4} , a designated directed cycle of 24 labels has nonzero square differences in $\mathbb{F}_p$, and each of its edge pairs is uniform among pairs with such a difference. This exact pair law supplies the arithmetic input below. The full reflection-positive law and its normalized nonexceptional part also satisfy conditional mixing, which makes inputs with large Fourier denominators negligible (Lemma 4.3). Section 3 constructs these laws from the graph estimates of Section 2.

The approximation errors require control of signed lifts beyond their means. Every fixed moment of G grows more slowly than any fixed positive power of N , uniformly in the prime set and in every retained family of complete denominator classes. Lemma 5.2 proves this by a conditional square-function estimate and symmetrization. The same bounds, with an explicit fiber-size cost, remain available after specifying some prime coordinates. Combined with conductor decay, they justify the Fourier truncations used in both estimates that follow.

For the change of scale, fixing prime coordinates whose product is at most a sufficiently small fixed power of N makes the low Fourier coefficients exact averages on an integer residue class modulo an integer D . Splitting that class into progressions of common difference D^2 and rescaling preserves square-difference avoidance: a square difference in a rescaled set would give a square difference in A . The seminorm’s square-affine symmetries therefore compare the specialized functional with the same functional at length $\lceil N/D^2 \rceil$, up to a factor tending to one and a power-decaying error (Proposition 7.2). This comparison applies to all specified residues, without using the exceptional splitting.

The arithmetic cancellation uses the normalized nonexceptional prime laws. Set aside the finitely many primes below the tuple-law threshold, together with modulus 8, in one fixed modulus M_0 . Independently choose one residue modulo M_0 for each slot. A fixed positive proportion of these assignments has strict square differences along the designated cycle: they are 1 modulo 8 and nonzero squares modulo every odd prime dividing M_0 . For such an assignment, divide $[N]$ into ordered intervals. Unless all cycle slots choose the same interval, one directed edge goes from an earlier interval to a later one. Truncating and expanding the other slots reduces its contribution to

the selected pair law. A signed kernel supported on actual shifts m^2 then bounds that contribution using the absence of integer square differences (Propositions 6.1 and 8.2). The event that all cycle slots choose one interval is itself small.

Finally, Section 9 expands the exceptional submeasures by inclusion–exclusion. Exceptional prime sets of large product are bounded directly by moment estimates. A smaller exceptional prime set with product q_B leads to the shorter scale approximately $N/(M_0 q_B)^2$. Its mass remains summable even after multiplication by q_B ; this absorbs the factor $q_B^{2a_0}$ incurred by a sufficiently small proposed decay exponent a_0 . Combining this bound with the universal scale comparison and the cancellation on a positive proportion of small-residue assignments gives a strict weighted recurrence. Induction proves $Y(N, A) \ll N^{-a_0}$, and the density inequality above gives Theorem 1.1 with $c = a_0/d$. Figure 1 records the principal inputs to this assembly.

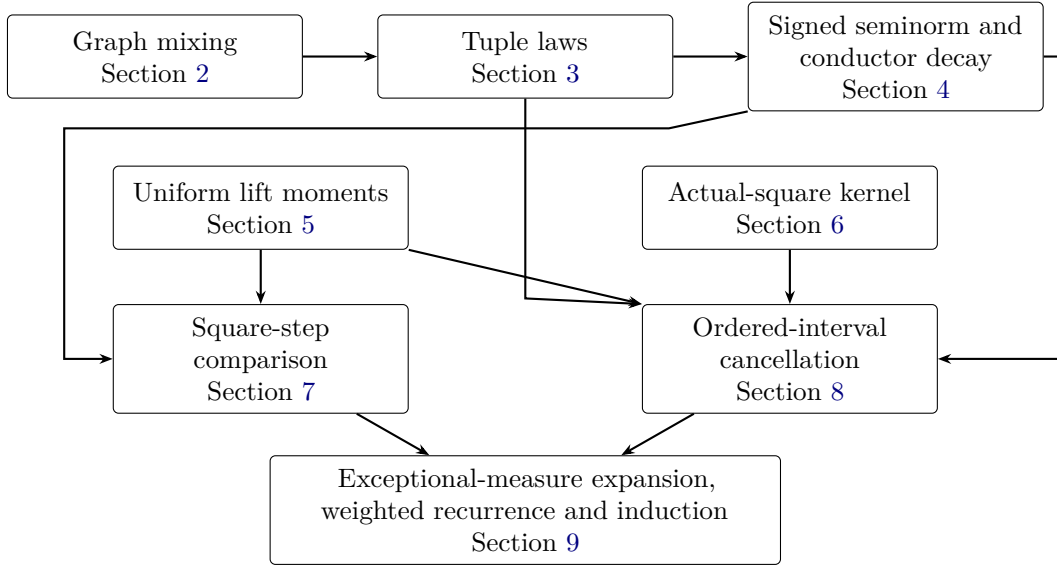


Figure 1: Principal proof inputs. The full tuple law supplies reflection positivity for the signed seminorm; the normalized nonexceptional law supplies the strict-square pair law for cancellation. Both have the conditional mixing used in conductor decay. The two lower branches are combined only in the weighted recurrence. Arrows indicate dependencies, not equivalences.

1.3 Conventions

All averages on finite sets are with respect to uniform probability measure unless another measure is displayed. All L^r norms of functions on $X_{\mathcal{P}}$ use its uniform product measure. Empty products are 1; the product over an empty prime set is a one-point space. We write $e(t) = \exp(2\pi it)$ and identify rational frequencies modulo 1. The *conductor* of a rational frequency is its reduced positive denominator; the conductor of 0 is 1.

For nonnegative quantities, $U \ll V$ means $U \leq CV$ for a constant with the dependencies stated in context. A uniform bound $U = N^{o(1)}$ means that for every $\varepsilon > 0$ there is a constant C_ε such that $U \leq C_\varepsilon N^\varepsilon$ for all allowed data. Fixed tuple sizes and fixed moment orders may affect C_ε ; sets, intervals, chosen residues and retained frequency families may not. All final choices of parameters are made independently of N and A .

2 Finite-field square constraints

This section controls systems of square constraints on independent field variables. A single constraint is close to a constant kernel in operator norm. Replacing constraints one at a time then counts any fixed directed graph without loops or opposite edges. Applying this count to a four-cycle of two lists gives the operator estimate needed for the tuple measures.

Throughout, finite sets carry uniform probability measure, and $e(t) = \exp(2\pi it)$. For an odd prime p , write

$$Q_p^* = \{u^2 : u \in \mathbb{F}_p^\times\}, \quad A_p(t) = 1_{Q_p^*}(t), \quad \delta_p = \frac{\sqrt{p} + 1}{2p} \leq p^{-1/2}.$$

For finite sets X, Y , the operator associated with a kernel $K : X \times Y \rightarrow \mathbb{R}$ is

$$T_K f(x) = \mathbb{E}_{y \in Y} K(x, y) f(y).$$

We use $\|K\|_{\text{op}}$ to mean the norm of T_K from $L^2(Y)$ to $L^2(X)$. A constant kernel c therefore denotes the operator $f \mapsto c\mathbb{E}f$.

Lemma 2.1 (Counting directed square constraints). *Let p be an odd prime and let $G = (W, E)$ be a finite directed graph with no loops and at most one edge on each unordered pair of distinct vertices. Put $m = |E|$, and let $(z_w)_{w \in W}$ be independent uniform elements of $\mathbb{F}_p$. Then*

$$\left| \mathbb{E} \prod_{(u,v) \in E} A_p(z_v - z_u) - 2^{-m} \right| \leq m\delta_p.$$

The assertion includes the empty graph, whose product is 1.

Proof. We first show that the kernel $b_p(x, y) = A_p(y - x) - 1/2$ has operator norm at most δ_p . The characters $x \mapsto e(kx/p)$, $k \in \mathbb{F}_p$, are an orthonormal eigenbasis. The constant eigenvalue is

$$\frac{|Q_p^*|}{p} - \frac{1}{2} = -\frac{1}{2p}.$$

For $k \neq 0$, put $G_p(k) = \sum_{u \in \mathbb{F}_p} e(ku^2/p)$. The corresponding eigenvalue is

$$\frac{1}{p} \sum_{t \in Q_p^*} e(kt/p) = \frac{G_p(k) - 1}{2p}.$$

Here the subtraction of 1 removes the zero square, and division by 2 removes the multiplicity of every nonzero square. Since $(u, v) \mapsto (u - v, u + v)$ is a bijection on $\mathbb{F}_p^2$,

$$|G_p(k)|^2 = \sum_{u, v \in \mathbb{F}_p} e(k(u^2 - v^2)/p) = \sum_{a, b \in \mathbb{F}_p} e(kab/p) = p.$$

Thus all eigenvalues have absolute value at most δ_p . In particular,

$$\left| \mathbb{E}_{x, y} f(x) g(y) (A_p(y - x) - \tfrac{1}{2}) \right| \leq \delta_p \|f\|_2 \|g\|_2. \quad (2.1)$$

The same bound holds with $x - y$ in place of $y - x$, since the transposed operator has the same norm.

Now replace the m edge indicators one at a time by $1/2$. In a telescoping error for an edge (u, v) , fix every variable except z_u, z_v . No other edge has both these endpoints. Consequently all the other factors separate as $Cf(z_u)g(z_v)$, where $|C|, |f|, |g| \leq 1$. Equation (2.1) bounds the conditional error by δ_p . Averaging the fixed variables and summing over the edges proves the assertion. $\square$

We next divide the variables into two independent lists. Only constraints between the lists enter the following kernel; constraints within either list must be treated separately.

Lemma 2.2 (Mixing between two lists). *Let I, J be finite disjoint sets, and let G be a directed graph on $I \sqcup J$ whose m edges all join I to J , with at most one edge on each unordered pair. The directions may vary from edge to edge. For $x \in \mathbb{F}_p^I$, $y \in \mathbb{F}_p^J$, write $z_i = x_i$ for $i \in I$, $z_j = y_j$ for $j \in J$, and set*

$$K(x, y) = \prod_{(u,v) \in E(G)} A_p(z_v - z_u).$$

Then

$$\|K - 2^{-m}\|_{\text{op}} \leq (32m)^{1/4} p^{-1/8}. \quad (2.2)$$

For $m = 0$, both sides are zero. In particular, the implied constant for any fixed pair of lists is independent of p .

Proof. Put $c = 2^{-m}$ and $B = K - c$. With respect to the normalized point-mass bases, T_B has matrix entries $B(x, y) / \sqrt{|\mathbb{F}_p^I| |\mathbb{F}_p^J|}$. If s_r are its singular values, then

$$\begin{aligned} \|B\|_{\text{op}}^4 &\leq \sum_r s_r^4 = \text{tr}((T_B T_B^*)^2) \\ &= \mathbb{E}_{x^0, x^1, y^0, y^1} B(x^0, y^0) B(x^0, y^1) B(x^1, y^0) B(x^1, y^1), \end{aligned} \quad (2.3)$$

where the four lists are independent and uniform. The kernel is real, so no conjugates appear in the last expression.

Let $\Omega = \{0, 1\}^2$. Expanding the last line gives

$$\sum_{F \subseteq \Omega} (-c)^{4-|F|} \mathbb{E} \prod_{(a,b) \in F} K(x^a, y^b).$$

For each F , its product is the constraint indicator of a graph on

$$(I \times \{0, 1\}) \sqcup (J \times \{0, 1\}).$$

Every base edge between $i \in I$ and $j \in J$ is copied, in its original direction, between (i, a) and (j, b) for every $(a, b) \in F$. The two endpoints determine the base unordered pair and the corner (a, b) . Hence this graph has no loops or repeated unordered edges, and has exactly $m|F|$ edges. Lemma 2.1 therefore gives

$$\mathbb{E} \prod_{(a,b) \in F} K(x^a, y^b) = c^{|F|} + E_F, \quad |E_F| \leq m|F| \delta_p.$$

The main terms sum to $(c - c)^4 = 0$. The remaining terms have absolute value at most

$$m \delta_p \sum_{t=0}^4 \binom{4}{t} t c^{4-t} = 4m(1+c)^3 \delta_p \leq 32m p^{-1/2}.$$

Combining this with (2.3) and taking fourth roots proves (2.2). $\square$

3 Finite reflection-positive probability laws

We construct probability laws on a fixed set of field labels. Their nonexceptional parts enforce a directed cycle of square differences. The full laws also satisfy positivity across a family of reflection cuts. To obtain both properties, we add a finite hierarchy of measures in which successively more blocks of the index set are marked. A marked block provides a positive quadratic form that dominates the small negative part at the corresponding unmarked cut.

3.1 The index set and the required properties

Fix the integers

$$h = 24, \quad t_0 = 80, \quad \ell = 2t_0 + 1 = 161. \quad (3.1)$$

Let Π be the set of permutation words on $[h] = \{1, \dots, h\}$, viewed as bijections from positions to symbols. Write

$$K = h!, \quad r = K/2, \quad V = \Pi^\ell, \quad d = |V| = K^\ell, \quad n = d/2.$$

The coordinates of a vertex $v \in V$ are called its blocks. Fix a cycle c of length h on the positions. Composition on the right, $w \mapsto wc$, cyclically shifts a word. Define $v_0, \dots, v_{h-1} \in V$ by giving every block of v_j the word c^j , and put $v_h = v_0$.

A reflection is specified by a block $b \in [\ell]$ and two distinct symbols α, β . It acts on V by swapping these symbols on the left in block b ; denote this involution by θ . Its associated cut is

$$V_+ = \{v : \alpha \text{ precedes } \beta \text{ in block } b\}, \quad V_- = \theta V_+.$$

The reflection has no fixed vertices, and both halves have size n .

For a finite alphabet X , a measure ν on X^V defines a matrix whose row index is $(z_v)_{v \in V_+}$ and whose column index is $(z_{\theta v})_{v \in V_+}$. We say that ν is *reflection positive* if this matrix is symmetric positive semidefinite for every cut above. Equivalently, symmetry holds and

$$\int F((z_v)_{v \in V_+}) F((z_{\theta v})_{v \in V_+}) d\nu(z) \geq 0$$

for every real function $F : X^{V_+} \rightarrow \mathbb{R}$.

Recall that $Q_p^* = \{u^2 : u \in \mathbb{F}_p^\times\}$ for an odd prime p . The following proposition summarizes the construction.

Proposition 3.1 (Tuple laws). *There is an absolute integer P such that, for every prime $p \geq P$, there are a probability measure μ_p on $\mathbb{F}_p^V$ and a nonnegative submeasure $\varepsilon_p \leq \mu_p$ whose mass satisfies*

$$\eta_p = \varepsilon_p(\mathbb{F}_p^V) \leq p^{-4}.$$

Writing

$$\mu_p^\circ = \frac{\mu_p - \varepsilon_p}{1 - \eta_p},$$

the following properties hold.

1. The full law μ_p is reflection positive. Both μ_p and ε_p are invariant under simultaneous changes $z_v \mapsto qz_v + a$, where $a \in \mathbb{F}_p$ and $q \in Q_p^*$. Every single-label marginal of μ_p and μ_p° is uniform on $\mathbb{F}_p$.

2. On the support of μ_p° ,

$$z_{v_{j+1}} - z_{v_j} \in Q_p^* \quad (0 \leq j < h).$$

The marginal of each ordered pair $(z_{v_j}, z_{v_{j+1}})$ under μ_p° is uniform on $\{(x, y) : y - x \in Q_p^*\}$.

3. Put $\gamma = 1/64$. For either $\rho_p = \mu_p$ or $\rho_p = \mu_p^\circ$, every $v \in V$, and every $f : \mathbb{F}_p \rightarrow \mathbb{C}$ with uniform mean zero,

$$\|\mathbb{E}_{\rho_p}[f(z_v) \mid (z_w)_{w \neq v}]\|_2 \leq p^{-\gamma} \|f\|_2. \quad (3.2)$$

The norm on the left uses the other-label marginal of ρ_p ; the norm on the right uses uniform measure on $\mathbb{F}_p$.

The cycle length h and the total tuple size d are fixed before any prime or integer interval is chosen. Thus constants depending on this finite construction remain absolute throughout the proof.

The construction and reflection estimates are given below. Lemma 3.6 will establish the final assertion and the precise pair marginal.

3.2 Measures with marked blocks

Start with one independent uniform label at each vertex of V . Require a strict square difference from v to w whenever $w_a = v_a c$ in at least $t_0 + 1$ blocks. In particular these constraints include every edge of the designated cycle. Multiplying the uniform product measure by the indicator of all these constraints gives the leading component, an unnormalized measure. Its cut matrices need not be positive semidefinite; the graph estimates control their negative parts.

To compensate for those negative parts, we introduce components in which vertices that agree outside a set of marked blocks draw their labels from a common list. A positive weight couples their choices of list entries. Marking a block then provides a lower bound at its reflection cuts, while leaving enough unmarked blocks to retain the cycle constraints until the last level. The following definitions implement this construction.

For $S \subseteq [\ell]$ with $s = |S| \leq t_0 + 1$, regard the blocks in S as marked; blocks outside S are called *clean*. We call the marking nonterminal if $s \leq t_0$ and terminal if $s = t_0 + 1$. The terminal components will form the exceptional measure. Let

$$U_S = \Pi^{[\ell] \setminus S}, \quad m_s = r^s,$$

and let $u_S : V \rightarrow U_S$ forget the marked blocks. At every address $u \in U_S$, choose a list $(Y_{u,i})_{i \in [m_s]}$ of independent uniform elements of $\mathbb{F}_p$. All lists are independent. The total number of list entries is

$$|U_S| m_s = K^{\ell-s} r^s = d/2^s \leq d. \quad (3.3)$$

For $s \leq t_0$, draw an address edge $u \rightarrow u'$ exactly when

$$\#\{a \notin S : u'_a = u_a c\} \geq \ell - t_0 = t_0 + 1. \quad (3.4)$$

For each such edge require

$$Y_{u',i'} - Y_{u,i} \in Q_p^* \quad \text{for all } i, i' \in [m_s]. \quad (3.5)$$

Let $I_S(Y)$ be the indicator of all these conditions. For $s = t_0 + 1$, impose no conditions and set $I_S = 1$. The graph on list entries defined by (3.5) will be denoted by $\mathcal{G}_S$.

There are no loops in the address graph: a permutation word cannot equal its nontrivial position shift. Nor can both directions occur between two addresses. Indeed, the clean blocks counted for the two directions are disjoint, since a block counted for both would imply $c^2 = 1$. Their combined required size is $2(t_0 + 1) > \ell$, which is impossible. Each unordered pair of list entries therefore supports at most one directed edge in $\mathcal{G}_S$. Lemma 2.1 applies to this graph and all its induced subgraphs.

Set

$$\delta = \frac{1}{10h^2}.$$

For distinct $v, w \in V$, define a symmetric interaction H_{vw}^S as follows. It is zero unless v, w differ by a symbol transposition in exactly one marked block. In that case, if the positions of the swapped

symbols have gap g , put $H_{vw}^S = \delta^g$. Independently of the lists, choose all indices $j_v \in [m_s]$ uniformly and independently, and give their joint distribution the unnormalized weight

$$W_S(j) = \exp \left(\sum_{\{v,w\} \subseteq V} H_{vw}^S 1_{\{j_v=j_w\}} \right). \quad (3.6)$$

Every sum over $\{v, w\}$ is over unordered pairs of distinct vertices, so each interaction is counted once.

Define the finite measure $\nu_{p,S}$ by the identity

$$\int G(z) d\nu_{p,S}(z) = \mathbb{E}_Y \mathbb{E}_j I_S(Y) W_S(j) G((Y_{u_S(v), j_v})_{v \in V}) \quad (3.7)$$

for all functions G on $\mathbb{F}_p^V$. Both expectations on the right are uniform probability averages. In particular, $\nu_{p,S}$ has not been normalized to have mass one. We also write $\nu(1)$ for the total mass of a finite measure ν .

Each vertex has at most $s \binom{h}{2}$ neighbors with nonzero interaction, and each interaction is at most δ . Thus, with the fixed constant

$$B = \exp \left(\frac{d(t_0 + 1) \binom{h}{2} \delta}{2} \right), \quad (3.8)$$

we have

$$1 \leq W_S(j) \leq B, \quad \nu_{p,S}(\mathbb{F}_p^V) \leq B. \quad (3.9)$$

For a cut with reflection θ , define the quadratic form

$$Q_{p,S}^b(F) = \int F((z_v)_{v \in V_+}) F((z_{\theta v})_{v \in V_+}) d\nu_{p,S}(z). \quad (3.10)$$

The notation suppresses the two symbols that specify the cut. Every component has a symmetric matrix at every cut, as we now verify.

Lemma 3.2 (Symmetry of the component matrices). *For every allowed mark set S and every reflection θ , the measure $\nu_{p,S}$ is invariant under the permutation $(z_v) \mapsto (z_{\theta v})$. Its matrix at that cut is symmetric.*

Proof. The array H^S is invariant under simultaneous application of θ to both endpoints. In the reflected block, conjugating a symbol transposition changes only the names of the swapped symbols, not their positional gap. Interactions in other blocks retain their gap as well. Hence relabeling indices by $j_v \mapsto j_{\theta v}$ preserves their uniform law and weight.

If the reflected block is marked, $u_S(\theta v) = u_S(v)$, so no relabeling of lists is needed. If it is unmarked, θ acts on addresses. Left symbol permutations commute with the right position shift c ; therefore it preserves the directed address relation (3.4), including its orientation. Relabeling lists by this address involution preserves their uniform law and I_S . In either case these bijections in (3.7) send the output label at v to the label at θv , proving invariance. At the cut this invariance exchanges the two matrix indices, which is symmetry. $\square$

3.3 Strict positivity at a marked cut

The next estimate is the source of domination between successive mark sets. It must remain valid when different index assignments produce identical field labels.

Lemma 3.3 (Marked-cut lower bound). *Put*

$$\kappa = \frac{9}{10}\delta^{h-1}, \quad \lambda = (e^\kappa - 1)^n, \quad c_* = \lambda m_{t_0+1}^{-n} > 0.$$

For every S with $|S| \leq t_0 + 1$, every cut in a marked block $b \in S$, and every real $F : \mathbb{F}_p^{V_+} \rightarrow \mathbb{R}$,

$$Q_{p,S}^b(F) \geq \lambda m_s^{-n} \mathbb{E}_{Y,j_+} I_S(Y) F((Y_{u_S(v),j_v})_{v \in V_+})^2. \quad (3.11)$$

Here the expectation uses all uniform lists and only the uniform indices on V_+ . In particular, the coefficient in (3.11) is at least c_* , independently of the prime and of the field-list values.

Proof. Index an $n \times n$ matrix by $x, y \in V_+$ and set

$$A_{xy} = H_{x,\theta y}^S.$$

The invariance of H^S used in Lemma 3.2, together with its symmetry, gives $A_{xy} = A_{yx}$.

Suppose the two cut symbols occupy positions $i < j$ in the reflected block of x , and set $g = j - i$. The diagonal entry is $A_{xx} = \delta^g$. A transposition taking x across the cut must act in this block and involve a cut symbol. Apart from swapping the two cut symbols themselves, it must either move the first symbol from i to a position $k > j$, or move the second from j to a position $k < i$. Its gap is therefore at least $g + 1$. There are fewer than h^2 such transpositions, so

$$\sum_{y \neq x} |A_{xy}| \leq h^2 \delta^{g+1} = \frac{1}{10} A_{xx}.$$

For every real vector a , using $2|a_x a_y| \leq a_x^2 + a_y^2$ gives

$$\begin{aligned} a^T A a &\geq \sum_x \left(A_{xx} - \sum_{y \neq x} |A_{xy}| \right) a_x^2 \\ &\geq \frac{9}{10} \delta^{h-1} \sum_x a_x^2 = \kappa \sum_x a_x^2. \end{aligned} \quad (3.12)$$

Write $m = m_s$ and $\mathcal{J} = [m]^{V_+}$. For $j \in \mathcal{J}$, define a vector with one nonzero coordinate for each vertex $x \in V_+$ by

$$u(j) = (1_{\{j_x=i\}})_{x \in V_+, i \in [m]}.$$

Using the reflected indices on the minus side, the cross factor of (3.6) is the matrix

$$C(j, k) = \exp(u(j)^T (A \otimes I_m) u(k)).$$

Each unordered cross pair is uniquely $\{x, \theta y\}$ with $x, y \in V_+$, so no extra factor of two appears in this formula.

By (3.12), $D = (A - \kappa I_n) \otimes I_m$ is positive semidefinite. The matrix

$$R(j, k) = \exp(u(j)^T D u(k))$$

is positive semidefinite: after taking a square root of D , the bilinear form is an inner product, and each term of the exponential series is a Gram matrix of tensor powers with a nonnegative coefficient. The convergent sum is positive semidefinite. Moreover $R(j, j) \geq 1$.

The remaining factor is

$$E(j, k) = \exp \left(\kappa \sum_{x \in V_+} 1_{\{j_x = k_x\}} \right).$$

As a matrix this is

$$E = \bigotimes_{x \in V_+} ((e^\kappa - 1)I_m + \mathbf{1}\mathbf{1}^T).$$

Every factor dominates $(e^\kappa - 1)I_m$, also for $m = 1$. It follows that E dominates $\lambda I_{\mathcal{J}}$. The entrywise product of positive semidefinite matrices is positive semidefinite: if they are Gram matrices, the products of their entries are the Gram matrix of pairwise tensor products. Denoting entrywise multiplication by $\circ$, we therefore obtain

$$C = (E - \lambda I_{\mathcal{J}}) \circ R + \lambda (I_{\mathcal{J}} \circ R) \succeq \lambda I_{\mathcal{J}}. \quad (3.13)$$

The internal weight on the plus half is

$$a(j) = \exp \left(\sum_{\{x, y\} \subseteq V_+} H_{xy}^S 1_{\{j_x = j_y\}} \right) \geq 1.$$

The reflected minus weight is the same function of its indices. Thus the full index matrix is $\text{diag}(a)C\text{diag}(a)$, which still dominates $\lambda I_{\mathcal{J}}$ by (3.13).

Fix the lists. Since $b \in S$, the two reflected vertices $v, \theta v$ have the same address. Consequently both sides of the quadratic form use the same vector

$$f_Y(j) = F((Y_{u_S(v), j_v})_{v \in V_+}).$$

This vector may have repeated entries. No injectivity of the map from index assignments to field assignments is required for the matrix lower bound. Since $|\mathcal{J}| = m^n$, the full uniform index average is bounded below by

$$\lambda m^{-2n} \sum_{j \in \mathcal{J}} f_Y(j)^2 = \lambda m^{-n} \mathbb{E}_{j \in \mathcal{J}} f_Y(j)^2.$$

Multiply by $I_S(Y)$ and average over the lists. This proves (3.11). The stated uniform coefficient follows from $m_s \leq m_{t_0+1}$. $\square$

3.4 The error at an unmarked cut

At an unmarked cut, the lists themselves split into two independent families. Lemma 2.2 controls the possible negative part of the cut form.

Lemma 3.4 (Unmarked-cut error). *Let $b \notin S$ specify a cut and suppose $s = |S| \leq t_0$. Let Y_+ consist of the lists at plus addresses, and let $I_+(Y_+)$ impose only the constraints internal to those addresses. For a real half-label function F , put*

$$D_{p,S}^b(F) = \mathbb{E}_{Y_+, j_+} I_+(Y_+) F((Y_{u_S(v), j_v})_{v \in V_+})^2.$$

Then, with the fixed constant $C_* = (32d^2)^{1/4} B^2$,

$$Q_{p,S}^b(F) \geq -C_* p^{-1/8} D_{p,S}^b(F). \quad (3.14)$$

If instead $s = t_0 + 1$, the form $Q_{p,S}^b$ is nonnegative.

Proof. Use θ to identify the minus-address list space with a second copy of the plus-address space. Symmetry of the directed graph gives the factorization

$$I_S(Y_+, Y_-) = I_+(Y_+)I_+(Y_-)K_b(Y_+, Y_-),$$

where K_b is the product of the constraints crossing the cut. Its graph is bipartite, simple on unordered pairs, and may contain both orientations on different pairs. If it has e_b edges, then $e_b \leq d^2$ by (3.3). Lemma 2.2 yields

$$\|K_b - 2^{-e_b}\|_{\text{op}} \leq (32d^2)^{1/4}p^{-1/8}.$$

Every nonzero index interaction changes a marked block, and so preserves the unmarked word in block b . No such interaction crosses the cut. The index weight therefore factors as $W_+(j_+)W_+(j_-)$, with the same function on the reflected halves and $1 \leq W_+ \leq B$. Set

$$G(Y_+) = I_+(Y_+)\mathbb{E}_{j_+}W_+(j_+)F((Y_{u_S(v),j_v})_{v \in V_+}).$$

Then $Q_{p,S}^b(F) = \langle G, T_{K_b}G \rangle$. The constant kernel contributes the nonnegative quantity $2^{-e_b}(\mathbb{E}G)^2$. Cauchy–Schwarz in the uniform index average, followed by $I_+^2 = I_+$ and $W_+ \leq B$, gives

$$\|G\|_2^2 \leq B^2 D_{p,S}^b(F).$$

The operator error proves (3.14). For $s = t_0 + 1$, no field constraints are present, so the same factorization gives exactly $Q_{p,S}^b(F) = (\mathbb{E}G)^2 \geq 0$. $\square$

3.5 Domination and the reflection-positive mixture

We now compare the two preceding lemmas. Marking a further block combines all its plus-side lists into one longer list. The precise length $m_s = r^s$ ensures that this identification preserves the uniform product distribution.

Lemma 3.5 (Domination after marking a block). *Let $b \notin S$, $|S| \leq t_0$, and set $S' = S \cup \{b\}$. For the cut in block b and every real half-label function F ,*

$$Q_{p,S'}^b(F) \geq c_* r^{-n} D_{p,S}^b(F). \quad (3.15)$$

Consequently the measure $\nu_{p,S} + p^{-1/16}\nu_{p,S'}$ has a positive semidefinite matrix at this cut whenever

$$p \geq P_{\text{ref}} := \max\{3, (C_* r^n / c_*)^{16}\}. \quad (3.16)$$

Proof. Let $\Pi_+ \subseteq \Pi$ consist of words in which the first cut symbol precedes the second. It has $r = K/2$ elements. Above each address in $U_{S'}$, the old plus addresses are indexed by $w \in \Pi_+$. Identify the new list of length $m_{s+1} = r m_s$ with the entries indexed by $(w, i) \in \Pi_+ \times [m_s]$ in those old lists. Applied at every new address, this identifies the entire new list collection with the old plus list collection; both have the same uniform product law. Figure 2 separates this relabeling from the index restriction used below.

If $s + 1 \leq t_0$, any new address edge already has at least $t_0 + 1$ matching blocks outside b . Every pair of old plus addresses above its endpoints therefore satisfies the old address edge condition, regardless of their block words in b . Old internal validity imposes the square constraint on all entries of every such pair of lists, and hence implies every new validity condition. If $s = t_0$, the new

$$\begin{aligned}
& \underbrace{\left((Y_{(u,w),i})_{i \in [m_s]} \right)_{w \in \Pi_+}}_{r \text{ lists, each of length } m_s} \longleftrightarrow \underbrace{(Y'_{u,(w,i)})_{(w,i) \in \Pi_+ \times [m_s]}}_{\text{one list of length } m_{s+1} = r m_s} \\
& j'_v = (w, i) \text{ uniform in } \Pi_+ \times [m_s] \xrightarrow{\text{restrict to } w=v_b} Y'_{u,(v_b,i)} = Y_{(u,v_b),i}, \\
& \mathbb{P}(w = v_b) = r^{-1}, \quad \mathbb{P}(\text{all } v \in V_+ \text{ select their own sublist}) = r^{-n}.
\end{aligned}$$

Figure 2: The two steps when marking a block b . For a new address $u \in U_{S \cup \{b\}}$, the old plus addresses above it are (u, w) , where $w \in \Pi_+$ is the word in block b ; the prime on Y' marks only a relabeling of the same entries. Regrouping the independent lists preserves their distribution without a loss. For each $v \in V_+$, with $u = u_{S \cup \{b\}}(v)$, restricting its independent new index to $w = v_b$ recovers its old emitted label. This second step has the total probability r^{-n} .

component is terminal and imposes no conditions, so the same implication holds. Thus under this identification

$$I_+(Y) \leq I_{S'}(Y).$$

Apply Lemma 3.3 to S' . In the expectation on its right-hand side, restrict every new half index j'_v to the sublist labeled by the actual word $v_b \in \Pi_+$. There are n independent indices, each selecting that sublist with probability $1/r$. The restriction therefore has probability exactly r^{-n} . Conditional on it, the residual indices in $[m_s]$ remain independent uniform variables, and the emitted half labels are exactly the old half labels. Since the test function is squared, restricting the expectation and then using $I_+ \leq I_{S'}$ gives

$$Q_{p,S'}^b(F) \geq c_* \mathbb{E}_{Y,j'_+} I_{S'}(Y) F(z_+)^2 \geq c_* r^{-n} D_{p,S}^b(F).$$

This is (3.15).

By Lemma 3.4, the sum of the two cut forms is at least

$$(c_* r^{-n} p^{-1/16} - C_* p^{-1/8}) D_{p,S}^b(F).$$

The bracket is nonnegative under (3.16). Symmetry follows from Lemma 3.2, completing the proof. $\square$

Define the unnormalized mixture and its mass by

$$M_p = \sum_{\substack{S \subseteq [\ell] \\ |S| \leq t_0+1}} p^{-|S|/16} \nu_{p,S}, \quad Z_p = M_p(\mathbb{F}_p^V). \quad (3.17)$$

For any fixed cut block b , it has the exact decomposition

$$\begin{aligned}
M_p &= \sum_{\substack{b \notin S \\ |S| \leq t_0}} p^{-|S|/16} (\nu_{p,S} + p^{-1/16} \nu_{p,S \cup \{b\}}) \\
&\quad + p^{-(t_0+1)/16} \sum_{\substack{b \notin S \\ |S| = t_0+1}} \nu_{p,S}.
\end{aligned} \quad (3.18)$$

Every component containing b appears exactly once, paired with the component obtained by removing b . Every component not containing b , except those of size $t_0 + 1$, is the first member of

one pair. The remaining components are terminal and are positive semidefinite at this unmarked cut by Lemma 3.4. Thus Lemma 3.5 and (3.18) show that M_p is reflection positive for $p \geq P_{\text{ref}}$. The threshold is uniform in the block, cut symbols, and mark sets.

3.6 Normalization, exceptional mass, and cycle support

Let e_0 be the number of edges of the leading list graph $\mathcal{G}_\emptyset$. For this component $m_0 = 1$ and $W_\emptyset = 1$. Lemma 2.1 gives

$$\left| \nu_{p,\emptyset}(\mathbb{F}_p^V) - 2^{-e_0} \right| \leq e_0 p^{-1/2}.$$

Put

$$c_0 = 2^{-e_0-1}, \quad P_{\text{mass}} = \max\{3, (2^{e_0+1}e_0)^2\}. \quad (3.19)$$

For every prime $p \geq P_{\text{mass}}$,

$$c_0 \leq \nu_{p,\emptyset}(\mathbb{F}_p^V) \leq Z_p \leq 2^\ell B. \quad (3.20)$$

The upper bound uses (3.9) and $p^{-|S|/16} \leq 1$.

We can therefore define

$$\mu_p = \frac{M_p}{Z_p}, \quad \varepsilon_p = \frac{p^{-(t_0+1)/16}}{Z_p} \sum_{|S|=t_0+1} \nu_{p,S}, \quad \eta_p = \varepsilon_p(\mathbb{F}_p^V). \quad (3.21)$$

The full law is a probability measure, and ε_p is a nonnegative submeasure. Moreover

$$\eta_p \leq C_{\text{exc}} p^{-81/16}, \quad C_{\text{exc}} = c_0^{-1} \binom{\ell}{81} B. \quad (3.22)$$

Since $81/16 - 4 = 17/16$, taking $p \geq C_{\text{exc}}^{16/17}$ ensures $\eta_p \leq p^{-4} < 1$. This defines the normalized nonexceptional law $\mu_p^\circ = (\mu_p - \varepsilon_p)/(1 - \eta_p)$. Positive normalization preserves reflection positivity of μ_p .

For later use, in either the full mixture or the normalized nonexceptional mixture, the total probability of components with $S \neq \emptyset$ is at most

$$\frac{2^\ell B}{c_0} p^{-1/16}. \quad (3.23)$$

Indeed their unnormalized weights are at most $p^{-1/16} \nu_{p,S}(\mathbb{F}_p^V)$, and the relevant total unnormalized mass is at least the leading mass c_0 in both cases.

Each component is invariant under simultaneous square-affine changes of field labels. To see this directly, apply $Y_{u,i} \mapsto qY_{u,i} + a$ to every list entry, where $q \in Q_p^*$. This bijection preserves the uniform list law, every strict-square difference condition, and the index weights. The output labels undergo the same affine map. Invariance therefore holds for μ_p , ε_p , and μ_p° . Translations act transitively on $\mathbb{F}_p$, so every single-label marginal of each positive-mass normalized component, and of each of these normalized mixtures, is uniform.

For $|S| \leq t_0$, the designated vertices v_j, v_{j+1} have clean words related by c in all $\ell - |S| \geq t_0 + 1$ clean blocks. Their addresses therefore form an edge in (3.4). Its conditions apply to every pair of list entries, including the two chosen by the indices. It follows that every nonterminal component is supported on

$$z_{v_{j+1}} - z_{v_j} \in Q_p^* \quad \text{for all } 0 \leq j < h. \quad (3.24)$$

The same support assertion holds for μ_p° .

The construction has now supplied the probability normalization, all reflection inequalities, affine invariance, the exceptional-mass bound, and the cycle support. It remains to verify conditional mixing and identify the selected pair marginals. Those facts use only the graph estimates, component masses, and invariance just established.

3.7 Conditional mixing and the selected pair marginals

The leading component of the construction has one independent field variable at each vertex before the constraints are imposed. We first show that, in this component, knowing all other labels predicts little about a mean-zero function of one label. The other components have small total mass, which permits the same conclusion with a weaker fixed exponent. We also identify the selected pair marginals that will supply the square-difference kernel.

Lemma 3.6 (Conditional mixing and pair marginals). *For the fixed index set V and the measures constructed above, there is an absolute threshold P such that the following statements hold for every prime $p \geq P$. Both probability measures*

$$\mu_p, \quad \mu_p^\circ = \frac{\mu_p - \varepsilon_p}{1 - \eta_p}, \quad \eta_p = \varepsilon_p(1),$$

have uniform marginals in each individual label. If ρ is either measure, $v \in V$, and λ is its marginal on $\mathbb{F}_p^{V \setminus \{v\}}$, then every function $f : \mathbb{F}_p \rightarrow \mathbb{C}$ with $\mathbb{E}_x f(x) = 0$ satisfies

$$\|\mathbb{E}_\rho[f(z_v) \mid (z_w)_{w \neq v}]\|_{L^2(\lambda)} \leq p^{-1/64} \|f\|_{L^2(\mathbb{F}_p)}. \quad (3.25)$$

The threshold is uniform in the choice of v and ρ .

For every selected cycle edge $v_j \rightarrow v_{j+1}$, its pair marginal under μ_p° is uniform on

$$\Omega_p = \{(x, y) \in \mathbb{F}_p^2 : y - x \in Q_p^*\}.$$

Its pair marginal under μ_p is

$$(1 - \eta_p) \text{Unif}(\Omega_p) + \eta_p \text{Unif}(\mathbb{F}_p^2). \quad (3.26)$$

Proof. Choose a fixed threshold P_0 beyond which the construction and its component mass bounds hold. By (3.20), the leading mass is at least $c_0 = 2^{-e_0-1}$. The affine invariance proved above gives uniform single-label marginals in every normalized positive-mass component. Thus a function with uniform mean zero is centered in each component separately, and its input L^2 norm is the same in each component.

Fix v . Write $x = z_v$ and $y = (z_w)_{w \neq v}$, and use independent uniform measures in these variables until otherwise stated. Since $m_0 = 1$, the density of $\nu_{p, \emptyset}$ is $L(y)K(x, y)$, where K is the product of the incident edge indicators and L is the product of all other edge indicators. Both are indicator functions. If m_v is the number of incident edges, put

$$c_v = 2^{-m_v}, \quad a(y) = \mathbb{E}_x K(x, y), \quad b_f(y) = \mathbb{E}_x f(x) K(x, y).$$

For $m_v > 0$, Lemma 2.2, with its operator transposed if necessary, gives

$$\|a - c_v\|_2 \leq \Delta_v, \quad \|b_f\|_2 \leq \Delta_v \|f\|_2, \quad \Delta_v = (32m_v)^{1/4} p^{-1/8}. \quad (3.27)$$

The incident constraints form a cross-list graph between the single variable x and the other labels. Incoming and outgoing edges are both allowed by that lemma. The first inequality applies the

centered kernel to the constant function, while the second uses $\mathbb{E}f = 0$. All norms in (3.27) use the uniform base measures.

Put $Z_0 = \nu_{p,\emptyset}(1)$, and let T_0 be the conditional operator for the normalized leading component. Its other-label marginal is $Z_0^{-1}L(y)a(y) dy$, and therefore

$$\|T_0 f\|_2^2 = Z_0^{-1} \mathbb{E}_y L(y) \frac{|b_f(y)|^2}{a(y)}. \quad (3.28)$$

The ratio is defined as zero where $a(y) = 0$, a set of zero marginal mass. On $\{a \geq c_v/2\}$, its contribution to (3.28) is at most $2c_0^{-1}c_v^{-1}\Delta_v^2\|f\|_2^2$. On the complementary set, Chebyshev's inequality gives

$$\mathbb{P}_y(a < c_v/2) \leq 4\Delta_v^2/c_v^2.$$

Weighted Cauchy–Schwarz gives the pointwise bound

$$|b_f(y)|^2 \leq a(y) \mathbb{E}_x |f(x)|^2 K(x, y), \quad \frac{|b_f(y)|^2}{a(y)} \leq \|f\|_2^2,$$

again with the zero-ratio convention. Thus no lower bound for $a(y)$ is needed on this small set. Since $L \leq 1$, its contribution is at most $4c_0^{-1}c_v^{-2}\Delta_v^2\|f\|_2^2$. We conclude that

$$\|T_0 f\|_2^2 \leq c_0^{-1}(2/c_v + 4/c_v^2)\Delta_v^2\|f\|_2^2 \leq C_{\text{lead}}p^{-1/4}\|f\|_2^2,$$

where the fixed constant

$$C_{\text{lead}} = c_0^{-1}(2^{e_0+1} + 2^{2e_0+2})\sqrt{32e_0}$$

works for every slot, since $m_v \leq e_0$. If $m_v = 0$, the label is independent of all others in this component, so $T_0 f = 0$ and the same bound holds.

Now let ρ be either μ_p or μ_p° . Represent it as a mixture of the normalized positive-mass components:

$$\rho = \sum_S \alpha_S \rho_S, \quad \rho_S = \frac{\nu_{p,S}}{\nu_{p,S}(1)}.$$

The components with zero mass are omitted. In the nonexceptional law, the terminal components $|S| = t_0 + 1$ are also omitted. In either case α_S is $p^{-|S|/16}\nu_{p,S}(1)$ divided by the sum of these masses over the retained components. By (3.23),

$$\sum_{S \neq \emptyset} \alpha_S \leq C_{\text{mix}}p^{-1/16}, \quad C_{\text{mix}} = 2^\ell B/c_0. \quad (3.29)$$

Introduce the component index S as a latent random variable and condition on it as well as the other labels. Conditional Jensen and the tower property imply

$$\begin{aligned} \|\mathbb{E}_\rho[f(z_v) \mid y]\|_2^2 &\leq \mathbb{E}_\rho \|\mathbb{E}[f(z_v) \mid y, S]\|_2^2 \\ &= \sum_S \alpha_S \|T_S f\|_2^2 \\ &\leq (C_{\text{lead}}p^{-1/4} + C_{\text{mix}}p^{-1/16})\|f\|_2^2. \end{aligned}$$

For the nonleading components we used only conditional-expectation contraction. Its input norm is the same uniform L^2 norm in each component, by the marginal statement proved above. Set $C_{\text{cond}} = C_{\text{lead}} + C_{\text{mix}}$. Enlarging the fixed prime threshold so that $P \geq \max\{P_0, 3, C_{\text{cond}}^{32}\}$ ensures

$$C_{\text{cond}}p^{-1/16} \leq p^{-1/32} \quad (p \geq P).$$

Taking square roots proves (3.25). All constants involved depend only on the fixed finite construction. In particular, this is one threshold for both laws and all slots, with no residual multiplicative constant in the operator estimate.

It remains to identify the pair marginals. By (3.24), the μ_p° -pair is supported on Ω_p . The square-affine group acts transitively on this set: the map taking (x, y) to (x', y') has

$$q = \frac{y' - x'}{y - x} \in Q_p^*, \quad a = x' - qx.$$

The affine invariance already proved forces the probability of every pair in Ω_p to equal $2/[p(p-1)]$.

In a terminal component, there are still $\ell - (t_0 + 1) = 80$ clean blocks. The two selected vertices have distinct clean addresses, and there are no field constraints. Conditional on every index choice, their labels are therefore selected from two independent uniform lists and form an independent uniform pair. The index weight is independent of the field entries, so this remains true after averaging the indices. The normalized exceptional mixture consequently has pair law $\text{Unif}(\mathbb{F}_p^2)$. Combining it with the nonexceptional part gives (3.26). $\square$

Equivalently, a selected pair under μ_p° has the form $(X, X + U)$, where X is uniform on $\mathbb{F}_p$ and U is independently uniform on Q_p^* . This exact description will be used only for the normalized nonexceptional law. The full law has the additional term displayed in (3.26); its conditional mixing estimate is nevertheless the same.

Proof of Proposition 3.1. Choose one integer P at least P_{ref} , P_{mass} , $C_{\text{exc}}^{16/17}$, and the threshold in Lemma 3.6. All these numbers depend only on the fixed parameters (3.1). Equations (3.18)–(3.21) give the normalized reflection-positive measure and its exceptional submeasure. The invariance and uniform marginals were proved above; (3.22) and (3.24) give the exceptional-mass bound and cycle support. Lemma 3.6 supplies the selected pair laws and (3.2). This proves all assertions. $\square$

4 Signed multilinear forms and conductor decay

The tuple laws provide two estimates for functions on products of prime fields. The first follows the reflection and chessboard method of [7, Theorem 2.2] and [6, Theorem 4.1]; the finite proof below verifies the required inequality for this particular tuple law. Reflection positivity gives a Hölder inequality for signed inputs and a seminorm that dominates the uniform mean. The conditional mixing estimate gives a separate bound for inputs supported on large Fourier denominators. This second estimate also applies to the nonexceptional laws.

Retain the index set $V = \Pi^\ell$, where Π consists of the permutation words on $[h]$, $h = 24$, and $\ell = 161$, and put $d = |V|$. In particular, d is even. Fix the prime threshold P from Proposition 3.1 and Lemma 3.6, and put $\gamma = 1/64$. For a finite set $\mathcal{P}$ of primes at least P , define

$$X_{\mathcal{P}} = \prod_{p \in \mathcal{P}} \mathbb{F}_p, \quad \mu_{\mathcal{P}} = \bigotimes_{p \in \mathcal{P}} \mu_p.$$

We regard $\mu_{\mathcal{P}}$ as a probability law on $X_{\mathcal{P}}^V$ by collecting, for each slot v , its labels at all primes into $z_v \in X_{\mathcal{P}}$. Every single-slot marginal is uniform on $X_{\mathcal{P}}$. Unless another measure is specified, norms and expectations on $X_{\mathcal{P}}$ use uniform probability measure. Set

$$\mathcal{Z}_{\mathcal{P}}((g_v)_{v \in V}) = \int \prod_{v \in V} g_v(z_v) d\mu_{\mathcal{P}}(z), \quad \mathcal{Z}_{\mathcal{P}}(g) = \mathcal{Z}_{\mathcal{P}}((g)_{v \in V}). \quad (4.1)$$

For an empty prime set, $X_{\emptyset}$ is a singleton and the product law has mass one; thus $\mathcal{Z}_{\emptyset}(g) = g^d$.

4.1 A finite signed reflection inequality

Proposition 4.1 (Signed Hölder inequality). *For every finite prime set $\mathcal{P}$ as above and real functions $g, g_v : X_{\mathcal{P}} \rightarrow \mathbb{R}$, one has*

$$Z_{\mathcal{P}}(g) \geq 0, \quad |\mathcal{Z}_{\mathcal{P}}((g_v)_{v \in V})| \leq \prod_{v \in V} Z_{\mathcal{P}}(g_v)^{1/d}. \quad (4.2)$$

The inequality includes inputs with zero diagonal integral.

Proof. Across any reflection cut, the matrix of $\mu_{\mathcal{P}}$ is the tensor product of the corresponding matrices of the μ_p . It is therefore symmetric positive semidefinite. This also holds for the empty prime set. We first record precisely what a reflection does to an assignment of functions.

Fix a block and two distinct symbols. Let θ exchange those symbols in that block, and let V_+ be the vertices in which the first symbol occurs before the second. Put $V_- = \theta V_+$. Define the two fold maps $\phi_+, \phi_- : V \rightarrow V$ by

$$\phi_+(v) = \begin{cases} v, & v \in V_+, \\ \theta v, & v \in V_-, \end{cases} \quad \phi_-(v) = \begin{cases} \theta v, & v \in V_+, \\ v, & v \in V_-. \end{cases}$$

For an assignment $a : V \rightarrow \mathbb{R}^{X_{\mathcal{P}}}$, abbreviate $\mathcal{Z}_{\mathcal{P}}((a(v))_{v \in V})$ by $\Lambda(a)$. The bilinear form of the cut matrix is evaluated on the two real functions

$$F(x) = \prod_{v \in V_+} a(v)(x_v), \quad G(x) = \prod_{v \in V_+} a(\theta v)(x_v).$$

Its mixed value is $\Lambda(a)$; its two quadratic values are $\Lambda(a \circ \phi_+)$ and $\Lambda(a \circ \phi_-)$. Cauchy–Schwarz for this positive semidefinite form gives

$$|\Lambda(a)|^2 \leq \Lambda(a \circ \phi_+) \Lambda(a \circ \phi_-), \quad \Lambda(a \circ \phi_{\pm}) \geq 0. \quad (4.3)$$

This remains valid when the form has a nontrivial null space. For a constant assignment, $F = G$; hence $Z_{\mathcal{P}}(g) \geq 0$.

We next specify a finite sequence of folds whose composition is constant. For a word w in block b , write $p_i = w^{-1}(i)$ for the position of symbol i . Let $\phi_{b,i}$ be the fold onto $p_i < p_{i+1}$, where $1 \leq i < h$. Swapping symbols $i, i+1$ on the left exchanges precisely the two entries p_i, p_{i+1} . Thus this fold compares these adjacent entries and puts them in increasing order.

In each block use the fixed sequence of comparisons

$$(1, 2), (2, 3), \dots, (h-1, h),$$

repeated $h-1$ times. One sweep moves the largest entry to the last position. Once the last k entries are the largest k entries in increasing order, the next sweep leaves them there and puts the largest remaining entry immediately before them. Induction shows that these sweeps sort every position array to $(1, \dots, h)$. Concatenating the sequences for all ℓ blocks gives fold maps $\phi_1, \dots, \phi_T$, with $T = \ell(h-1)^2$, such that

$$\phi_T \circ \dots \circ \phi_1(v) = v_* \quad \text{for every } v \in V, \quad (4.4)$$

where v_* has the identity word in every block.

To use this sequence, fix any finite nonempty list of real functions $f_1, \dots, f_k$ and set

$$M = \max_{a: V \rightarrow [k]} \left| \mathcal{Z}_{\mathcal{P}}((f_{a(v)})_{v \in V}) \right|.$$

The maximum ranges over all assignments, with repetition, and exists because there are only k^d assignments. If $M > 0$, choose a maximizer. Each of its folded assignments still takes values in the same finite list, so each folded integral is at most M in absolute value. Equation (4.3) says that those two integrals are nonnegative and their product is at least M^2 . They must therefore both equal M . In particular, either fold of any maximizing assignment is again maximizing.

Apply the folds in reverse sorting order, $\phi_T, \phi_{T-1}, \dots, \phi_1$. Pullbacks compose on the right, so the resulting assignment is $a \circ \phi_T \circ \dots \circ \phi_1$, which is constant by (4.4). Every intermediate integral is M , and the last is one of the diagonal values $Z_{\mathcal{P}}(f_i)$. Conversely, every constant assignment was included in the maximum. It follows that

$$M = \max_{1 \leq i \leq k} Z_{\mathcal{P}}(f_i). \quad (4.5)$$

When $M = 0$, all these diagonal values are zero, so the same identity holds. This is a finite argument and requires no limiting sequence of folds. The maximizing-assignment argument is analogous to the proof of the abstract chessboard estimate in [6, Theorem 4.1].

Finally, for $\epsilon > 0$, apply (4.5) to the list

$$f_v = \frac{g_v}{(Z_{\mathcal{P}}(g_v) + \epsilon)^{1/d}}, \quad v \in V.$$

Every denominator is positive and $Z_{\mathcal{P}}(f_v) \leq 1$. Multilinearity consequently gives

$$|Z_{\mathcal{P}}((g_v)_{v \in V})| \leq \prod_{v \in V} (Z_{\mathcal{P}}(g_v) + \epsilon)^{1/d}.$$

Letting $\epsilon \downarrow 0$ proves (4.2), including zero diagonal values. □

4.2 The seminorm and its symmetries

Proposition 4.1 allows us to define, for real g ,

$$\|g\|_{\mathcal{P}} = Z_{\mathcal{P}}(g)^{1/d}.$$

For $B \subseteq \mathcal{P}$, let $\mathbb{E}_B g$ denote uniform averaging over the coordinates in B , holding the other coordinates fixed. We may regard this either as a function on $X_{\mathcal{P}}$ independent of B , or as a function on $X_{\mathcal{P} \setminus B}$.

Lemma 4.2 (Seminorm properties). *The map $g \mapsto \|g\|_{\mathcal{P}}$ is a seminorm on the real functions on $X_{\mathcal{P}}$, and*

$$|\mathbb{E} g| \leq \|g\|_{\mathcal{P}}. \quad (4.6)$$

For $a_p \in \mathbb{F}_p$ and $r_p \in \mathbb{Q}_p^*$, let $A(x)_p = r_p x_p + a_p$. Then

$$\|g \circ A\|_{\mathcal{P}} = \|g\|_{\mathcal{P}}. \quad (4.7)$$

For every $B \subseteq \mathcal{P}$,

$$\|\mathbb{E}_B g\|_{\mathcal{P} \setminus B} = \|\mathbb{E}_B g\|_{\mathcal{P}} \leq \|g\|_{\mathcal{P}}. \quad (4.8)$$

Proof. Nonnegativity is already proved. Since d is even, homogeneity of the diagonal form gives $\|cg\|_{\mathcal{P}} = |c| \|g\|_{\mathcal{P}}$ for every real c . For real functions g_1, g_2 , expand $Z_{\mathcal{P}}(g_1 + g_2)$ multilinearly and apply (4.2) to each term. With $a = \|g_1\|_{\mathcal{P}}$ and $b = \|g_2\|_{\mathcal{P}}$, this gives

$$0 \leq Z_{\mathcal{P}}(g_1 + g_2) \leq \sum_{S \subseteq V} a^{|S|} b^{d-|S|} = (a + b)^d.$$

Taking the nonnegative d -th root proves the triangle inequality, also when one of the seminorms vanishes.

The probability normalization gives $Z_{\mathcal{P}}(1) = 1$. Insert g in one slot and 1 in all others in (4.2). The uniform single-slot marginal identifies the left side with $|\mathbb{E}g|$, proving (4.6). The simultaneous affine invariance of each μ_p proves (4.7); its parameters may be chosen independently at different primes.

Let G_B be the additive subgroup of $X_{\mathcal{P}}$ consisting of vectors supported on B . Then

$$(\mathbb{E}_B g)(x) = \frac{1}{|G_B|} \sum_{t \in G_B} g(x+t).$$

The triangle inequality and translation invariance show that this average has seminorm at most $\|g\|_{\mathcal{P}}$. Since the averaged function is independent of B , integration over the prime laws in B , each of mass one, gives

$$Z_{\mathcal{P}}(\mathbb{E}_B g) = Z_{\mathcal{P} \setminus B}(\mathbb{E}_B g).$$

This proves (4.8). All statements include the empty prime set, for which the seminorm is absolute value. $\square$

4.3 Conductor decay

The characters of $X_{\mathcal{P}}$ are indexed by $\xi \in \mathbb{Q}/\mathbb{Z}$ whose reduced denominator $q(\xi)$ divides $\prod_{p \in \mathcal{P}} p$, with $q(0) = 1$. We write such a character as $e(z\xi)$: it means $e(n\xi)$ for any integer n satisfying $n \equiv z_p \pmod{p}$ for all $p \in \mathcal{P}$. This is well defined by the Chinese remainder theorem. The Fourier coefficient of g is

$$\widehat{g}(\xi) = \mathbb{E}_{z \in X_{\mathcal{P}}} g(z) e(-z\xi).$$

For $U \subseteq \mathcal{P}$, put $q_U = \prod_{p \in U} p$, including $q_{\emptyset} = 1$. The exact prime support of a character is the set of primes at which it is nontrivial; its denominator is exactly the product of these primes.

Recall the exceptional mass and the normalized nonexceptional law,

$$\eta_p = \varepsilon_p(\mathbb{F}_p^V), \quad \mu_p^{\circ} = \frac{\mu_p - \varepsilon_p}{1 - \eta_p}.$$

Proposition 3.1 gives $\eta_p \leq p^{-4} < 1$.

Lemma 4.3 (Conductor decay). *Let $\mathcal{P}$ be a finite set of primes at least P , and at each prime choose ρ_p to be either μ_p or μ_p° . Let $R \geq 1$, and let $g_w : X_{\mathcal{P}} \rightarrow \mathbb{C}$, $w \in V$, be functions. Suppose that for one slot v , $\widehat{g}_v(\xi) = 0$ whenever $q(\xi) \leq R$. Then*

$$\left| \int \prod_{w \in V} g_w(z_w) d\bigotimes_{p \in \mathcal{P}} \rho_p(z) \right| \leq R^{-\gamma} \|g_v\|_2 \prod_{w \neq v} \|g_w\|_{2(d-1)}. \quad (4.9)$$

All norms on the right use uniform probability measure on $X_{\mathcal{P}}$. The constant in the inequality is one.

Proof. Fix the distinguished slot v . For each p , let $Y_p = \mathbb{F}_p^{V \setminus \{v\}}$, let λ_p be the marginal law of the other labels under ρ_p , and define

$$T_p f(y) = \mathbb{E}_{\rho_p}[f(z_v) \mid (z_w)_{w \neq v} = y].$$

This is an operator from $L^2(\mathbb{F}_p, \text{unif})$ to $L^2(Y_p, \lambda_p)$; its values on marginal-null points are irrelevant. It preserves constants, and the tower property gives

$$T_p 1 = 1, \quad \int T_p f d\lambda_p = \mathbb{E}_{x \in \mathbb{F}_p} f(x). \quad (4.10)$$

For uniform-mean-zero f , Lemma 3.6 gives

$$\|T_p f\|_{L^2(\lambda_p)} \leq p^{-\gamma} \|f\|_{L^2(\text{unif})}. \quad (4.11)$$

Let H_p^0 be the mean-zero subspace of $L^2(\mathbb{F}_p, \text{unif})$, and let K_p^0 be the mean-zero subspace of $L^2(Y_p, \lambda_p)$. Thus

$$L^2(\mathbb{F}_p, \text{unif}) = \mathbb{C}1 \oplus H_p^0, \quad L^2(Y_p, \lambda_p) = \mathbb{C}1 \oplus K_p^0,$$

with orthogonal sums. Equation (4.10) says that $T_p H_p^0 \subseteq K_p^0$. Notice that centering in the codomain refers to λ_p , which need not be uniform.

For $U \subseteq \mathcal{P}$, let H_U be the tensor subspace with factor H_p^0 when $p \in U$ and factor $\mathbb{C}1$ otherwise; define K_U in the same way using K_p^0 . The H_U form an orthogonal direct sum of $L^2(X_{\mathcal{P}}, \text{unif})$. Distinct K_U, K_W are also orthogonal: at a prime in the symmetric difference of U, W , one tensor factor is constant and the other has mean zero. Inner products of pure tensors therefore vanish, and linearity gives the assertion for the whole subspaces.

Under the product law $\bigotimes_p \rho_p$, the marginal of all the other labels is $\lambda = \bigotimes_p \lambda_p$. Conditional independence across primes shows that the operator from the distinguished slot to all other labels is

$$T = \bigotimes_{p \in \mathcal{P}} T_p.$$

For completeness, on an input $\prod_p f_p(z_{v,p})$, the conditional expectation is $\prod_p T_p f_p$, since the conditional law is a product over primes. Such pure tensors span the finite-dimensional domain, proving the operator identity. Consequently

$$T(H_U) \subseteq K_U, \quad \|T|_{H_U}\| \leq \prod_{p \in U} p^{-\gamma} = q_U^{-\gamma}. \quad (4.12)$$

The norm bound follows by applying (4.11) one tensor factor at a time; on the constant factors the norm is one.

Nontrivial characters at p form an orthonormal basis for H_p^0 . Thus H_U consists precisely of Fourier sums with exact denominator q_U . Decompose the distinguished input as $g_v = \sum_{q_U > R} g_{v,U}$, with $g_{v,U} \in H_U$. The images of distinct summands are orthogonal by (4.12) and the orthogonality of the K_U . Hence

$$\begin{aligned} \|T g_v\|_{L^2(\lambda)}^2 &= \sum_{q_U > R} \|T g_{v,U}\|_{L^2(\lambda)}^2 \\ &\leq \sum_{q_U > R} q_U^{-2\gamma} \|g_{v,U}\|_2^2 \leq R^{-2\gamma} \|g_v\|_2^2. \end{aligned}$$

Only distinct exact supports need have orthogonal images; no such assertion is made for different characters within a single support.

Finally condition the multilinear integral on all labels except those in slot v , and apply Cauchy–Schwarz. Its absolute value is at most

$$R^{-\gamma} \|g_v\|_2 \left\| \prod_{w \neq v} g_w(z_w) \right\|_{L^2(\lambda)}.$$

Hölder's inequality with $d - 1$ factors and the uniform marginal in each remaining slot give

$$\left\| \prod_{w \neq v} g_w(z_w) \right\|_{L^2(\lambda)} \leq \prod_{w \neq v} \left(\mathbb{E}_\lambda |g_w(z_w)|^{2(d-1)} \right)^{1/(2(d-1))} = \prod_{w \neq v} \|g_w\|_{2(d-1)}.$$

This proves (4.9). If $\mathcal{P}$ is empty, the support hypothesis forces $g_v = 0$, so the assertion holds there too. No reflection positivity of the laws ρ_p was used. $\square$

5 Uniform moment bounds for Fourier lifts

We now transfer a bounded function on an integer interval to the product of its prime residue spaces. Keeping rational frequencies of bounded denominator need not preserve pointwise bounds or positivity. The replacement we require is a uniform bound for every fixed moment, valid also after retaining arbitrary complete denominator classes. We first control a square function of the exact prime supports, then recover the sum by symmetrization. For related arithmetic moment estimates and Fourier lifting, see [3, Theorem 2 of the arXiv version] and [9, Section 3]; the uniform estimate used here is proved below.

Let $\mathcal{P}$ be a finite set of primes, and give

$$X_{\mathcal{P}} = \prod_{p \in \mathcal{P}} \mathbb{F}_p$$

uniform probability measure. A character is indexed by $\xi = \sum_{p \in \mathcal{P}} a_p/p \pmod{1}$, where $a_p \in \{0, \dots, p-1\}$, and we write

$$e(z\xi) = \prod_{p \in \mathcal{P}} e(a_p z_p/p), \quad e(t) = \exp(2\pi i t).$$

Its exact prime support is $\{p : a_p \neq 0\}$, and its reduced denominator is the product of these primes. In particular $q(0) = 1$. Summing a geometric progression in each coordinate shows that these characters form an orthonormal basis. For $U \subseteq \mathcal{P}$, put $q_U = \prod_{p \in U} p$, with $q_\emptyset = 1$.

Let I be a nonempty consecutive interval of integers, of length L , and let $f : I \rightarrow \mathbb{C}$. Define

$$\widehat{f}_I(\xi) = \frac{1}{L} \sum_{n \in I} f(n) e(-n\xi), \quad f_U(z_U) = \sum_{q(\xi)=q_U} \widehat{f}_I(\xi) e(z_U \xi).$$

For $Q \geq 1$, the Fourier lift is

$$\mathcal{L}_{I, \mathcal{P}, Q} f = \sum_{\substack{U \subseteq \mathcal{P} \\ q_U \leq Q}} f_U.$$

If $\mathcal{F}$ is a subfamily of these supports, write $\mathcal{L}_{I, \mathcal{P}, Q}^{\mathcal{F}} f = \sum_{U \in \mathcal{F}} f_U$. Thus retaining a support means retaining all frequencies of its exact denominator. Each f_U depends only on z_U and has mean zero in each coordinate belonging to U . Real f gives real f_U , because the summation set is closed under negation.

Remark 5.1 (A lift need not be nonnegative). Take distinct primes p, q with $p, q \leq Q < pq$ and $\mathcal{P} = \{p, q\}$. Let I have length divisible by pq , and let f be the indicator of the simultaneous congruences $n \equiv a \pmod{p}$ and $n \equiv b \pmod{q}$. The cutoff retains the supports $\emptyset, \{p\}, \{q\}$ and omits $\{p, q\}$. Since each residue pair occurs $|I|/(pq)$ times, summing these three components gives

$$\mathcal{L}_{I, \mathcal{P}, Q} f(z_p, z_q) = \frac{1_{z_p=a}}{q} + \frac{1_{z_q=b}}{p} - \frac{1}{pq}.$$

In particular, the lift equals $-1/(pq)$ when both coordinates miss the prescribed residues. This is why the multilinear estimates must allow signed inputs even when the original function is an indicator.

Lemma 5.2 (Uniform moments of Fourier lifts). *Let $Q \geq 1$, let I have length $L \geq 10Q^6$, and suppose $\|f\|_\infty \leq M$. Let $\mathcal{F}$ be any subfamily of $\{U \subseteq \mathcal{P} : q_U \leq Q\}$, and put*

$$k = \max\{|U| : U \subseteq \mathcal{P}, q_U \leq Q\}.$$

For every integer $r \geq 1$, define

$$K_{2r} = \left(\frac{(2r)!}{2^r r!} \right)^{1/(2r)}, \quad A_r = 2eK_{2r} 2^{(r+3)/4}.$$

Then

$$\left\| \mathcal{L}_{I, \mathcal{P}, Q}^{\mathcal{F}} f \right\|_{2r} \leq 2M(k+1)A_r^k. \quad (5.1)$$

Consequently, for fixed r , this norm is at most $MQ^{o_r(1)}$ as $Q \rightarrow \infty$, uniformly in $I, f, \mathcal{P}$, and $\mathcal{F}$.

Proof. Conditional square-sum bound. We establish a conditional square-sum bound before estimating moments. For every $B \subseteq \mathcal{P}$ with $q_B \leq Q$, and every fixed value z_B , we claim that

$$\sum_{\substack{U \in \mathcal{F} \\ U \supseteq B}} \mathbb{E}(|f_U|^2 \mid z_B) \leq 4M^2 4^{|B|}. \quad (5.2)$$

The conditional expectation here integrates the coordinates outside B ; those outside U are irrelevant to its summand.

To compute its coefficients, write $U = B \sqcup A$ and uniquely decompose a frequency on U as $\xi = \xi_B + \eta$, with exact supports B and A , respectively. After fixing z_B , the coefficient of the outside character $e(z_A \eta)$ is

$$c_B(\eta; z_B) = \frac{1}{L} \sum_{n \in I} f(n) e(-n\eta) \prod_{p \in B} (p 1_{n \equiv z_p(p)} - 1). \quad (5.3)$$

Indeed, the sum over the nonzero frequencies at p is

$$\sum_{a=1}^{p-1} e(a(z_p - n)/p) = p 1_{n \equiv z_p(p)} - 1.$$

Let Ω_B denote the union of the outside-frequency sets arising from $U \in \mathcal{F}$ with $U \supseteq B$. Different outside supports give disjoint frequency sets, and all their denominators are at most Q/q_B and coprime to q_B . Conditional Parseval therefore identifies the left side of (5.2) with $\sum_{\eta \in \Omega_B} |c_B(\eta; z_B)|^2$.

Expand the product in (5.3) by subsets $J \subseteq B$. Apart from its sign, the corresponding coefficient vector is

$$c_J(\eta) = \frac{q_J}{L} \sum_{\substack{n \in I \\ n \equiv z_J(q_J)}} f(n) e(-n\eta), \quad \eta \in \Omega_B. \quad (5.4)$$

The congruence in this expression is the simultaneous congruence at the primes of J , or equivalently its unique class modulo q_J . For $J = \emptyset$ it imposes no restriction.

Write its progression as $n = a + q_J t$, where t ranges over a consecutive interval T of length L_J . Since $q_J \leq Q$ and $L \geq 10Q^6$,

$$|L_J - L/q_J| \leq 1, \quad L_J \geq 0.9L/q_J, \quad q_J L_J / L \leq 1.1.$$

After removing the phase $e(-a\eta)$, (5.4) is $q_J L_J / L$ times the normalized Fourier analysis of $t \mapsto f(a + q_J t)$ at frequencies $q_J \eta$.

These frequencies remain distinct. In fact the denominator of $\eta - \eta'$ is coprime to q_J , so $q_J(\eta - \eta') \in \mathbb{Z}$ implies $\eta = \eta' \pmod{1}$. Multiplication by q_J also preserves each reduced outside denominator. Thus all the resulting frequencies have denominators at most Q , are pairwise separated in circle distance by at least Q^{-2} , and number at most $\sum_{q \leq Q} \varphi(q) \leq Q^2$, where $\varphi(1) = 1$ accounts for the zero frequency.

This is a separated-frequency estimate of large-sieve type; compare [13, Theorem 1]. The length hypothesis permits the following elementary Gram-matrix bound. Consider the Gram matrix of these normalized exponential vectors in $L^2(T)$. Its diagonal is 1. For distinct frequencies θ, θ' , geometric summation gives

$$\left| \frac{1}{L_J} \sum_{t \in T} e(t(\theta - \theta')) \right| \leq \frac{1}{2L_J \|\theta - \theta'\|_{\mathbb{R}/\mathbb{Z}}} \leq \frac{Q^2}{2L_J}.$$

Its operator norm is at most its maximum absolute row sum, hence at most

$$1 + \frac{Q^4}{2L_J} \leq 1 + \frac{Q^4 q_J}{1.8L} \leq 1 + \frac{1}{18Q} \leq \frac{19}{18}.$$

The norm squared of the Fourier analysis map equals this Gram-matrix norm. Consequently

$$\left(\sum_{\eta \in \Omega_B} |c_J(\eta)|^2 \right)^{1/2} \leq 1.1 \sqrt{19/18} \left(\frac{1}{L_J} \sum_{t \in T} |f(a + q_J t)|^2 \right)^{1/2} \leq 2M.$$

The Euclidean triangle inequality over the $2^{|B|}$ subsets J proves (5.2).

Square-function moments. We next use this bound to control

$$S(z) = \left(\sum_{U \in \mathcal{F}} |f_U(z_U)|^2 \right)^{1/2}.$$

Put $T_r = \mathbb{E} S^{2r}$ and $T_0 = 1$. Expand the first $r - 1$ factors in this moment and fix their supports $U_1, \dots, U_{r-1}$. Their product depends only on coordinates in $W = U_1 \cup \dots \cup U_{r-1}$, and $|W| \leq (r-1)k$. For a last support U , let $B = U \cap W$. Integrating outside W gives exactly $\mathbb{E}(|f_U|^2 \mid z_B)$, because the coordinates of f_U outside B are fresh. For each fixed intersection B , its sum over $U \cap W = B$ is bounded by (5.2). Only B with $q_B \leq Q$ can occur, and there are at most $2^{(r-1)k}$ possible intersections. It follows that

$$T_r \leq 4M^2 4^k 2^{(r-1)k} T_{r-1}.$$

Induction gives

$$T_r \leq (4M^2)^r 4^{kr} 2^{kr(r-1)/2}, \quad \|S\|_{2r} \leq 2M 2^{k(r+3)/4}. \quad (5.5)$$

This conditioning argument permits arbitrary overlaps among supports. The remaining task is to recover their sum from this square function.

Recovering the sum. Fix a size $j \geq 1$ and color the coordinates independently and uniformly with j colors. A support is called transversal if it contains exactly one coordinate of each color. Each support of size j is transversal with probability $\pi_j = j!/j^j$. Fix one coloring c , and let $\mathcal{T}$ be its retained transversal supports in $\mathcal{F}$.

Take independent copies X_p^0, X_p^1 of every coordinate. Write swap_p for interchanging that pair, and define

$$D_U = \prod_{p \in U} (I - \text{swap}_p) f_U(X^0).$$

Coordinatewise centering gives $\mathbb{E}_{X^1} D_U = f_U(X^0)$: in the expanded product, every term using at least one copied coordinate has zero conditional mean. Jensen's inequality therefore gives

$$\left\| \sum_{U \in \mathcal{T}} f_U \right\|_{2r} \leq \left\| \sum_{U \in \mathcal{T}} D_U \right\|_{L^{2r}(X^0, X^1)}.$$

Swapping independent copies coordinate by coordinate is a standard symmetrization device; see [5, Section 4 of the arXiv version]. Independently swapping each coordinate pair preserves their joint law and multiplies D_U by the product of the associated signs. With independent random signs $(\epsilon_p)_{p \in \mathcal{P}}$,

$$\left\| \sum_{U \in \mathcal{T}} D_U \right\|_{L^{2r}(X^0, X^1)} = \left\| \sum_{U \in \mathcal{T}} D_U \prod_{p \in U} \epsilon_p \right\|_{L^{2r}(X^0, X^1, \epsilon)}. \quad (5.6)$$

The signs here are indexed by individual coordinates, not merely by colors. Each monomial contains one coordinate sign from each color class.

For completeness, the scalar random-sign bound with the constant K_{2r} above is

$$\left\| \sum_i a_i \epsilon_i \right\|_{2r} \leq K_{2r} \left(\sum_i |a_i|^2 \right)^{1/2}. \quad (5.7)$$

For nonnegative coefficients b_i , expansion of the even moment retains only multiplicities $2m_i$ with $\sum_i m_i = r$. Since $(2m)! \geq 2^m m!$, that moment is at most

$$\frac{(2r)!}{2^r} \sum_{\sum m_i = r} \prod_i \frac{b_i^{2m_i}}{m_i!} = \frac{(2r)!}{2^r r!} \left(\sum_i b_i^2 \right)^r.$$

For complex a_i , expand $(\sum a_i \epsilon_i)^r (\sum \bar{a}_i \epsilon_i)^r$ and take absolute values termwise; its surviving terms are bounded by the expansion with $b_i = |a_i|$. This proves (5.7) in the required generality.

For fixed copies X^0, X^1 , iterate (5.7) through the j color classes. At each stage one uses Minkowski in L^r , in the form

$$\left\| \left(\sum_i |B_i|^2 \right)^{1/2} \right\|_{2r} \leq \left(\sum_i \|B_i\|_{2r}^2 \right)^{1/2}.$$

Induction on the number of color classes therefore gives

$$\left\| \sum_{U \in \mathcal{T}} D_U \prod_{p \in U} \epsilon_p \right\|_{L^{2r}(\epsilon)} \leq K_{2r}^j \left(\sum_{U \in \mathcal{T}} |D_U|^2 \right)^{1/2}.$$

To estimate this difference square function, index the choices of copies by $\eta \in \{0, 1\}^j$, one choice for each color. Since each $U \in \mathcal{T}$ has one coordinate of every color,

$$D_U = \sum_{\eta \in \{0, 1\}^j} (-1)^{|\eta|} f_U((X_p^{\eta_{c(p)}})_{p \in U}).$$

For each fixed η , the selected full coordinate vector has the original product law. Minkowski in $L^{2r}(\ell^2(\mathcal{T}))$ consequently yields

$$\left\| \left(\sum_{U \in \mathcal{T}} |D_U|^2 \right)^{1/2} \right\|_{2r} \leq 2^j \left\| \left(\sum_{U \in \mathcal{T}} |f_U|^2 \right)^{1/2} \right\|_{2r} \leq 2^j \|S\|_{2r}.$$

Together with (5.6), this proves the transversal bound $(2K_{2r})^j \|S\|_{2r}$.

Average over colorings. Since every support is retained with probability π_j , another application of Minkowski gives

$$\left\| \sum_{\substack{U \in \mathcal{F} \\ |U|=j}} f_U \right\|_{2r} \leq \frac{j^j}{j!} (2K_{2r})^j \|S\|_{2r} \leq (2eK_{2r})^j \|S\|_{2r}.$$

Here $j! \geq (j/e)^j$, which also follows by integrating $\log x$ from 1 to j . The term of size zero is constant and bounded in absolute value by S . Summing over $0 \leq j \leq k$ and using (5.5) proves (5.1).

Finally, the product of k distinct primes is at least $(k+1)!$, so $(k+1)! \leq Q$. The bound $\log(k!) \geq k \log k - k$ gives $k = O(\log Q / \log \log Q)$. For fixed r , the logarithm of the right side of (5.1), after removing M , is therefore $o(\log Q)$. Every bound used above was independent of the interval's location, the ambient prime set, and the retained family. This proves the asserted uniformity. $\square$

We record separately the cost of fixing residue coordinates. This cost is a power of the size of the fixed fiber, and must remain explicit when the moment bound is used in multilinear estimates.

Lemma 5.3 (Specialization of lifts). *Let $B \subseteq \mathcal{P}$, let $z_B \in X_B$, and let $G : X_{\mathcal{P}} \rightarrow \mathbb{C}$. For every finite $p \geq 1$,*

$$\|G(z_B, \cdot)\|_{L^p(X_{\mathcal{P} \setminus B})} \leq q_B^{1/p} \|G\|_{L^p(X_{\mathcal{P}})}. \quad (5.8)$$

In the setting of Lemma 5.2, this gives, for every fixed z_B ,

$$\left\| (\mathcal{L}_{I, \mathcal{P}, Q}^{\mathcal{F}} f)(z_B, \cdot) \right\|_{2r} \leq 2M q_B^{1/(2r)} (k+1) A_r^k.$$

The same bound holds for either part obtained by cutting the remaining denominator at any $H \geq 1$: before specialization these parts correspond to the whole-support families

$$\mathcal{F}_{\leq H, B} = \{U \in \mathcal{F} : q_{U \setminus B} \leq H\}, \quad \mathcal{F}_{> H, B} = \{U \in \mathcal{F} : q_{U \setminus B} > H\}.$$

After specialization their Fourier supports lie, respectively, at remaining denominators at most H and greater than H .

Proof. The specified fiber has uniform probability $1/q_B$. Its contribution to $\mathbb{E}|G|^p$ is

$$q_B^{-1} \mathbb{E}_{z_{\mathcal{P} \setminus B}} |G(z_B, z_{\mathcal{P} \setminus B})|^p.$$

Discarding all other nonnegative contributions proves (5.8). Apply Lemma 5.2 to each displayed family. Every character of f_U , when specialized on B , retains exactly the coordinate support $U \setminus B$. Combining such characters can cancel coefficients but cannot create frequencies outside the asserted remaining-denominator ranges. This proves the last assertion. $\square$

In particular, for $d \geq 2$, a multilinear estimate using one L^2 norm and $d - 1$ norms in $L^{2(d-1)}$, with the same coordinates fixed in every slot, incurs the total specialization factor

$$q_B^{1/2} \left(q_B^{1/(2(d-1))} \right)^{d-1} = q_B.$$

Two specialized L^2 factors likewise incur q_B . Multiplying the original bounded function by a fixed constant simply multiplies every norm bound by the absolute value of that constant.

For later uniform estimates one can make the subpower dependence entirely explicit. Put

$$k_*(Q) = \max\{j \in \mathbb{Z}_{\geq 0} : (j+1)! \leq Q\}, \quad \mathfrak{M}_r(Q) = 2(k_*(Q) + 1)A_r^{k_*(Q)}.$$

The moment bound is at most $M\mathfrak{M}_r(Q)$, and for every fixed r and $\epsilon > 0$, one has $\mathfrak{M}_r(Q) \leq Q^\epsilon$ for sufficiently large Q . The threshold may depend on r, ϵ , but on none of the intervals, functions, fibers, or retained families. Thus any fixed finite product of the envelopes $\mathfrak{M}_r(Q)$ remains subpower, even when their fixed moment orders are very large. The displayed input bounds and specialization powers remain explicit.

6 A signed kernel for actual square differences

The preceding tuple measures impose square-difference constraints in residue fields. This section supplies the passage to actual positive integer squares. We construct a signed measure supported on such squares and approximate its Fourier transform uniformly by a sum over small rational frequencies. Because every actual square shift vanishes between the supports under consideration, the signs of the measure cause no difficulty. The weighted square sums and major/minor-arc analysis have a direct antecedent in [9, Section 5]. Here the signed divisor weight is chosen to match the strict-square pair law of our finite-field tuples.

For the remainder of the paper, fix

$$\beta = 10^{-5}. \tag{6.1}$$

Write $e(t) = \exp(2\pi it)$, and let $q(\xi)$ be the reduced denominator of $\xi \in \mathbb{Q}/\mathbb{Z}$, with $q(0) = 1$. For every rational frequency define the normalized unit quadratic sum

$$\mathfrak{g}(\xi) = \frac{1}{\varphi(q(\xi))} \sum_{m \in (\mathbb{Z}/q(\xi)\mathbb{Z})^\times} e(\xi m^2), \quad \mathfrak{g}(0) = 1. \tag{6.2}$$

Here φ is Euler's totient function. If $D = 8d_0$ with d_0 odd and squarefree, call a residue modulo D a *strict square class* if it is 1 modulo 8 and a nonzero square modulo every prime dividing d_0 .

Proposition 6.1 (Ordered square-kernel estimate). *Let $N \geq 1$ be an integer, put $H = N^{\beta/2}$, and let $D = 8d_0 \leq N^\beta$, where d_0 is odd and squarefree. Suppose $F, J : [N] \rightarrow \mathbb{C}$ satisfy $|F|, |J| \leq 1$ and are supported in residue classes a, b modulo D , respectively. Assume that $b - a$ is a strict square class, that $x < y$ whenever $x \in \text{supp } F$ and $y \in \text{supp } J$, and that no such difference $y - x$ is the square of a positive integer. With*

$$\widehat{F}(\xi) = \frac{1}{N} \sum_{n=1}^N F(n)e(-n\xi), \quad \widehat{J}(\xi) = \frac{1}{N} \sum_{n=1}^N J(n)e(-n\xi),$$

one has

$$D \left| \sum_{\substack{\xi \in \mathbb{Q}/\mathbb{Z}: q(\xi) \leq H \\ q(\xi) \text{ squarefree, } (q(\xi), D)=1}} \mathfrak{g}(\xi) \widehat{F}(-\xi) \widehat{J}(\xi) \right| \ll H^{-1/3}. \tag{6.3}$$

The implied constant is absolute and is independent of all the data in the statement.

Proof. Put $r = b - a$ modulo D , and denote the frequency set in (6.3) by Ξ . We first define the model and actual-square kernels. We then compare their Fourier transforms on minor and major arcs, and finish with the exact bilinear Fourier identity.

The two kernels. For every odd prime $p \mid d_0$, choose one root c_p of r modulo p . The conditions $m \equiv c_p \pmod{p}$ for all such primes, together with m odd, specify one residue class modulo $D' = 2d_0 = D/4$. Let ρ_D be D' times the indicator of this class. Thus ρ_D has periodic mean one and supremum D' . For every odd prime $p \nmid D$, set

$$w_p(m) = \frac{p \mathbf{1}_{p \mid m} - 1}{p - 1} = \frac{1}{p - 1} \sum_{j=1}^{p-1} e(jm/p), \quad w_u(m) = \prod_{p \mid u} w_p(m), \quad (6.4)$$

where u is squarefree and coprime to D , and $w_1 = 1$. The second equality follows by summing all characters modulo p . In particular $|w_u| \leq 1$.

The factor $1 - w_p(m) = p(p - 1)^{-1} \mathbf{1}_{p \nmid m}$ is the density of the uniform unit law modulo p relative to the uniform law. Hence for any positive integer q coprime to D ,

$$\sum_{u \mid \text{rad}(q)} (-1)^{\omega(u)} w_u(m) = \prod_{p \mid q} (1 - w_p(m)) = \frac{q}{\varphi(q)} \mathbf{1}_{(m, q) = 1}. \quad (6.5)$$

Here $\omega(u)$ counts the distinct prime divisors of u , and $\text{rad}(q)$ is the product of the distinct primes dividing q . Thus the complete divisor sum converts a uniform residue into a uniform unit, whose square has Fourier multiplier $\mathfrak{g}$. We truncate this sum at $u \leq H$ in the actual-square kernel. At a rational frequency with denominator $q \leq H$ coprime to D , all divisors of $\text{rad}(q)$ remain present; the periodic mean of every term containing a prime outside q vanishes. This will give the exact low-denominator comparison. The other rational coefficients will be bounded by quadratic Gauss sums.

Define two finitely supported functions on $\mathbb{Z}$ by

$$k_0(n) = \frac{D}{N} \mathbf{1}_{1 \leq n \leq N} \mathbf{1}_{n \equiv r \pmod{D}} \sum_{\xi \in \Xi} \mathfrak{g}(\xi) e(-\xi n), \quad (6.6)$$

$$k(n) = \sum_{1 \leq m \leq \sqrt{N}} \mathbf{1}_{n=m^2} \frac{2m}{N} \rho_D(m) \sum_{\substack{u \leq H \\ \text{squarefree} \\ (u, D) = 1}} (-1)^{\omega(u)} w_u(m). \quad (6.7)$$

The first kernel represents the rational expression in the proposition; the second is supported on actual positive integer squares. The factor $2m/N$ is the discrete counterpart of the substitution $dn = 2m dm$, matching the uniform weight $1/N$ in the model kernel. For these kernels use the positive-sign Fourier transforms

$$S_0(\alpha) = \sum_{n \in \mathbb{Z}} k_0(n) e(\alpha n), \quad S(\alpha) = \sum_{n \in \mathbb{Z}} k(n) e(\alpha n), \quad \alpha \in \mathbb{R}/\mathbb{Z}.$$

Our immediate goal is

$$\sup_{\alpha \in \mathbb{R}/\mathbb{Z}} |S(\alpha) - S_0(\alpha)| \ll H^{-1/3}. \quad (6.8)$$

Put $V_N(t) = N^{-1} \sum_{n=1}^N e(tn)$. Expanding the congruence indicator in (6.6) gives

$$S_0(\alpha) = \sum_{\xi \in \Xi} \sum_{j \bmod D} \mathfrak{g}(\xi) e(-jr/D) V_N(\alpha - \xi + j/D). \quad (6.9)$$

There are at most DH^2 terms, each of coefficient modulus at most one, and their centers have denominators at most DH . The centers are distinct. Indeed, an equality $\xi - j/D = \xi' - j'/D$ would make $\xi - \xi'$ have denominator both coprime to D and dividing D , so that $\xi = \xi'$ and $j = j'$ modulo D .

Every $\alpha_0 \in \mathbb{Q}/\mathbb{Z}$ has a unique decomposition $\alpha_0 = \alpha_D + \alpha_1$ in which the denominator of α_D has only prime factors dividing D and $q_1 = q(\alpha_1)$ is coprime to D . Thus the coefficient of the center α_0 in (6.9), taken to be zero when it is absent, is

$$c_0(\alpha_0) = \begin{cases} e(\alpha_D r) \mathfrak{g}(\alpha_1), & q(\alpha_D) \mid D, \quad q_1 \leq H, \quad q_1 \text{ squarefree}, \\ 0, & \text{otherwise.} \end{cases} \quad (6.10)$$

The sign is worth recording: in (6.9), $\alpha_D = -j/D$, and its coefficient contains $e(-jr/D)$.

Minor arcs. We initially assume N is sufficiently large by an absolute amount. Take the major arcs to be the closed balls of radius $2N^{-9/10}$ about all reduced rationals of denominator at most $N^{1/10}$. They are disjoint for sufficiently large N , since two distinct centers are at least $N^{-1/5}$ apart. The minor arcs are their complement. We write $\|t\|$ for the distance of a real number t to the nearest integer.

For α on the minor arcs, Dirichlet approximation with $Q_0 = \lfloor N^{9/10} \rfloor$ gives coprime integers a, q satisfying

$$1 \leq q \leq Q_0, \quad |\alpha - a/q| \leq \frac{1}{qQ_0} \leq \frac{1}{q^2},$$

where the difference is represented by a real number of least absolute value. For completeness, partitioning the circle into Q_0 intervals and placing the $Q_0 + 1$ points $0, \alpha, \dots, Q_0\alpha$ in them gives integers $1 \leq h \leq Q_0$ and b with $|h\alpha - b| \leq Q_0^{-1}$. Reducing b/h to a/q gives the displayed bounds. If $q \leq N^{1/10}$, the first approximation bound would put α within $2N^{-9/10}$ of a major-arc center. Consequently

$$N^{1/10} < q \leq N^{9/10}. \quad (6.11)$$

Let $M = \lfloor \sqrt{N} \rfloor$. Uniformly in real θ and integers $1 \leq L \leq M$, differencing and a geometric sum give

$$\left| \sum_{m=1}^L e(\alpha m^2 + \theta m) \right|^2 \ll M + \sum_{j=1}^M \min(M, \|2\alpha j\|^{-1}). \quad (6.12)$$

Indeed, pairs at distance j leave a linear phase with coefficient $2\alpha j$ in the inner sum. Its remaining factors, including $e(\theta j)$, have modulus one. This also proves the asserted uniformity in the linear twist.

Partition the j indices into consecutive blocks of length $\lfloor q/4 \rfloor$. Two distinct indices in a block differ by $0 < h < q/4$. The residue $2ah$ is nonzero modulo q : since $(a, q) = 1$, its vanishing would force $q \mid 2h$, whereas $0 < 2h < q/2$. This argument includes even q . Hence

$$\|2\alpha h\| \geq \|2ah/q\| - 2h/q^2 \geq \frac{1}{2q}.$$

In each block the points $2\alpha j$ are therefore separated on the circle by at least $1/(2q)$. Ordering their distances to zero bounds the contribution of one block by $O(M + q \log(2q))$: only a bounded number can lie within distance $1/q$ of zero, and the others contribute a harmonic sum. There are $O(M/q + 1)$ blocks, so

$$\sum_{j=1}^M \min(M, \|2\alpha j\|^{-1}) \ll (M/q + 1)(M + q \log(2q)). \quad (6.13)$$

By (6.11), (6.12)–(6.13) imply

$$\sup_{\theta \in \mathbb{R}} \max_{L \leq M} \left| \sum_{m=1}^L e(\alpha m^2 + \theta m) \right| \ll N^{9/20} \sqrt{\log N}.$$

Partial summation with the weight $2m/N$ gives

$$\sup_{\theta \in \mathbb{R}} \left| \sum_{m \leq \sqrt{N}} \frac{2m}{N} e(\alpha m^2 + \theta m) \right| \ll N^{-1/20} \sqrt{\log N}. \quad (6.14)$$

For a periodic function, use Fourier coefficients normalized by its period. The Fourier coefficient ℓ^1 norm of ρ_D is D' : its character expansion has D' coefficients of modulus one. By (6.4), the corresponding norm of each w_p is one, and that of w_u is at most one by the convolution inequality. The entire periodic weight multiplying the quadratic phase in S thus has Fourier coefficient ℓ^1 norm at most $D'[H] \leq DH$. Expansion into these characters and (6.14) yield

$$|S(\alpha)| \ll DHN^{-1/20} \sqrt{\log N} \quad (\alpha \text{ on the minor arcs}). \quad (6.15)$$

Every center in (6.9) has denominator at most $DH \leq N^{3\beta/2} < N^{1/10}$, so it is a major-arc center. On the minor arcs its distance from α exceeds $2N^{-9/10}$. The elementary estimate

$$|V_N(t)| \leq \min(1, (2N\|t\|)^{-1})$$

therefore gives

$$|S_0(\alpha)| \ll DH^2 N^{-1/10} \quad (\alpha \text{ on the minor arcs}). \quad (6.16)$$

Reduction on a major arc to periodic means. Write $\alpha = \alpha_0 + y$, where $q = q(\alpha_0) \leq N^{1/10}$ and $|y| \leq 2N^{-9/10}$. If $\alpha_c \neq \alpha_0$ is a center in (6.9), rational separation gives $\|\alpha_c - \alpha_0\| \geq 1/(qDH)$. Because $qDH \leq N^{1/10+3\beta/2}$, for all sufficiently large N we have $\|\alpha - \alpha_c\| \geq 1/(2qDH)$. The contributions of these other centers sum to

$$S_0(\alpha) = c_0(\alpha_0)V_N(y) + O(qD^2H^3/N). \quad (6.17)$$

For an admissible u , put

$$a_u(m) = \rho_D(m)w_u(m)e(\alpha_0 m^2), \quad A_u = \frac{1}{T_u} \sum_{m \bmod T_u} a_u(m), \quad T_u = \text{lcm}(q, D', u).$$

The sequence a_u has period $T_u \leq qDu$ and supremum at most D . Breaking a partial sum into complete periods and one remaining interval shows, for every real $t \geq 0$, that

$$\left| \sum_{1 \leq m \leq t} a_u(m) - A_u t \right| \ll DT_u.$$

This includes the fractional part of t . For $f_y(x) = (2x/N)e(yx^2)$ on $[0, \sqrt{N}]$, direct differentiation gives

$$|f_y(\sqrt{N})| + \int_0^{\sqrt{N}} |f'_y(x)| dx \ll \frac{1 + |y|N}{\sqrt{N}}.$$

Stieltjes partial summation consequently gives

$$\sum_{m \leq \sqrt{N}} \frac{2m}{N} e(y m^2) a_u(m) = A_u \int_0^{\sqrt{N}} \frac{2x}{N} e(y x^2) dx + O\left(\frac{q D^2 u (1 + |y|N)}{\sqrt{N}}\right).$$

The integral equals $N^{-1} \int_0^N e(y t) dt$. Comparing the integral to its integer Riemann sum bounds its difference from $V_N(y)$ by $O((1 + |y|N)/N)$. Since $|A_u| \leq D$, that error is dominated by the preceding displayed error. Summing over $u \leq H$ proves

$$S(\alpha) = c(\alpha_0) V_N(y) + O\left(\frac{q D^2 H^2 (1 + |y|N)}{\sqrt{N}}\right), \quad c(\alpha_0) = \sum_{\substack{u \leq H \\ \text{squarefree} \\ (u, D)=1}} (-1)^{\omega(u)} A_u. \quad (6.18)$$

The use of $\sqrt{N}$ as a real endpoint in this calculation accounts for rounding, while the sum itself contains only integers m with $m^2 \leq N$.

We have reduced the major-arc comparison to $c(\alpha_0)$ and $c_0(\alpha_0)$. We next compute the contribution at primes dividing D , and then the divisor sieve on the coprime denominator q_1 .

The prescribed roots, including the prime two. At an odd prime $p \mid D$, the local factor of ρ_D is $p \mathbf{1}_{m \equiv c_p \pmod{p}}$. For every $j \geq 1$, squaring maps the lifts of c_p modulo p^j bijectively to the residues congruent to r modulo p . Indeed, if x is such a lift, then

$$(x + t p^k)^2 \equiv x^2 + 2 x t p^k \pmod{p^{k+1}},$$

and $2x$ is invertible modulo p . This proves the bijection inductively over k . For a primitive additive character $e(am^2/p^j)$, its expectation under this normalized root restriction is therefore $e(ar/p)$ if $j = 1$, and zero if $j \geq 2$: in the latter case one averages a nontrivial character over all the lifts of a single residue modulo p . If no p -part is present in the character, the expectation is one.

At two the local factor of ρ_D is $2 \mathbf{1}_{m \text{ odd}}$. For $j \leq 3$, every odd square is r modulo 2^j , so the expectation of $e(am^2/2^j)$ is $e(ar/2^j)$. For $j \geq 3$, the kernel of the squaring homomorphism on $(\mathbb{Z}/2^j\mathbb{Z})^\times$ has exactly four elements. To see this, if $(x-1)(x+1)$ is divisible by 2^j , exactly one of these two consecutive even factors has 2-adic valuation one, and the other is divisible by 2^{j-1} . The four solutions modulo 2^j are $1, -1, 1 + 2^{j-1}, -1 + 2^{j-1}$. The image of squaring thus has 2^{j-3} elements. Every element of this image is 1 modulo 8, and this residue class also has 2^{j-3} elements. The square of a uniform odd residue modulo 2^j is consequently uniform among all residues equal to 1 modulo 8. For $j > 3$, a primitive character averages to zero on that class.

Apply these local facts to the decomposition $\alpha_0 = \alpha_D + \alpha_1$ already used in (6.10). Chinese remaindering shows that the factor at primes dividing D is

$$\mathbb{E}_m \rho_D(m) e(\alpha_D m^2) = \begin{cases} e(\alpha_D r), & q(\alpha_D) \mid D, \\ 0, & q(\alpha_D) \nmid D. \end{cases} \quad (6.19)$$

The expectation here is the uniform average over any common period. The normalized odd-root restrictions, and the normalized oddness restriction at two, are precisely responsible for the factor D in the model kernel. In particular, choosing only one odd root introduces no additional factor in (6.19).

The coprime divisor sieve. Every prime of an admissible u is coprime to D . If such a prime does not divide q_1 , its mean-zero factor w_p is independent of the exponential and all other factors by the Chinese remainder theorem. Thus $A_u = 0$ unless $u \mid \text{rad}(q_1)$.

Suppose first that $q_1 \leq H$. All divisors of $\text{rad}(q_1)$ then occur in the sum. By (6.5), the divisor weight is the density of the uniform unit law modulo q_1 relative to the uniform law. If p^j exactly divides q_1 with $j \geq 2$, its local primitive unit quadratic average vanishes. Indeed, conditional on a unit x modulo p^{j-1} , the p lifts $x + tp^{j-1}$ have phase factors differing by $e(2atx/p)$, where $p \nmid 2ax$. Their sum is zero. If q_1 is squarefree, the unit average is exactly $\mathbf{g}(\alpha_1)$. Together with (6.19), this proves

$$c(\alpha_0) = c_0(\alpha_0) \quad (q_1 \leq H). \quad (6.20)$$

For $q_1 > H$ the model coefficient is zero, and we need a bound for the truncated divisor sum. We give the prime-power calculation explicitly. For an odd prime p , an integer $j \geq 1$, and $p \nmid a$, write

$$G_j(a) = \frac{1}{p^j} \sum_{m \bmod p^j} e(am^2/p^j), \quad G_0(a) = 1.$$

Because two is invertible modulo p^j , the substitution $(m, n) \mapsto (m - n, m + n)$ is a bijection. Consequently

$$|G_j(a)|^2 = \frac{1}{p^{2j}} \sum_{u, v \bmod p^j} e(auv/p^j) = p^{-j}. \quad (6.21)$$

Let $T_j = \mathbb{E}_{m \bmod p^j} \mathbf{1}_{p|m} e(am^2/p^j)$. For $j = 1$ one has $T_1 = 1/p$; for $j \geq 2$, substituting $m = pt$ gives $T_j = p^{-1}G_{j-2}(a)$. It follows that

$$B_j := \mathbb{E}_{m \bmod p^j} w_p(m) e(am^2/p^j) = \frac{pT_j - G_j(a)}{p-1}, \quad |B_j| \leq 2p^{-j/2}. \quad (6.22)$$

For $j \geq 2$, the last bound follows from $(p+1)/(p-1) \leq 2$ and (6.21). For $j = 1$, the bound obtained from the formula is $(1 + p^{-1/2})/(p-1)$; its ratio to $p^{-1/2}$ is $1/(\sqrt{p}-1) \leq 2$. Thus the exponent-one case is included.

For a fixed divisor $u \mid \text{rad}(q_1)$, the coprime local mean is a product of factors from (6.21) and (6.22). Its modulus is at most $2^{\omega(u)} q_1^{-1/2}$. The factor at primes of D has modulus at most one by (6.19). Dropping the restriction $u \leq H$ after taking absolute values therefore gives

$$|c(\alpha_0)| \leq q_1^{-1/2} \sum_{u \mid \text{rad}(q_1)} 2^{\omega(u)} = 3^{\omega(q_1)} q_1^{-1/2} \ll q_1^{-1/3} \leq H^{-1/3}. \quad (6.23)$$

The penultimate implied constant is absolute. For example, one may use

$$C_* = \prod_{\substack{p < 729 \\ p \text{ odd prime}}} \max(1, 3p^{-1/6}),$$

since $3 \leq p^{1/6}$ for every prime $p \geq 729$. Repeated prime powers only increase $q_1^{1/6}$. This argument covers all prime powers in q_1 , and if $q(\alpha_D) \nmid D$, every coefficient under discussion is already zero by (6.19).

Completion of the uniform comparison. On the minor arcs, (6.15) and (6.16) apply. On a major arc, (6.17), (6.18), (6.20) and (6.23) apply, with $|V_N(y)| \leq 1$. For clarity, every approximation error is bounded here using the fixed value (6.1):

$$\begin{aligned} DHN^{-1/20}\sqrt{\log N} &\leq N^{-0.049985}\sqrt{\log N}, \\ DH^2N^{-1/10} &\leq N^{-0.09998}, \\ qD^2H^3/N &\leq N^{-0.899965}, \\ \frac{qD^2H^2(1+|y|N)}{\sqrt{N}} &\leq 3N^{-0.29997}. \end{aligned}$$

In the last two lines $q \leq N^{1/10}$ and $|y| \leq 2N^{-9/10}$. Each bound is $O(N^{-\beta/6}) = O(H^{-1/3})$ with an absolute constant. Together with the coefficient bound this proves (6.8) for all sufficiently large N . All size and separation thresholds used above are absolute. For the bounded remaining range, the elementary bounds $\|S\|_\infty \ll DH$ and $\|S_0\|_\infty \leq DH^2$ absorb that range by increasing the absolute constant. Thus (6.8) is uniform in all the data of the proposition.

The directional bilinear identity. It remains to pass from the Fourier comparison to the supports in the statement. Extend F and J by zero to $\mathbb{Z}$. For any finitely supported kernel h define

$$\mathcal{B}_h(F, J) = \frac{1}{N} \sum_{x \in \mathbb{Z}} \sum_{n \in \mathbb{Z}} h(n) F(x) J(x+n).$$

Both sums use counting measure; the displayed $1/N$ normalizes only the base variable. Let

$$\tilde{F}(\alpha) = \sum_{x \in \mathbb{Z}} F(x) e(-x\alpha), \quad \tilde{J}(\alpha) = \sum_{y \in \mathbb{Z}} J(y) e(-y\alpha), \quad S_h(\alpha) = \sum_{n \in \mathbb{Z}} h(n) e(\alpha n).$$

Expanding and integrating the characters imposes $y = x + n$, giving the bilinear identity

$$\mathcal{B}_h(F, J) = \frac{1}{N} \int_0^1 S_h(\alpha) \tilde{F}(-\alpha) \tilde{J}(\alpha) d\alpha.$$

There are no conjugates in this identity. Cauchy–Schwarz and counting-measure Plancherel now give

$$|\mathcal{B}_{k-k_0}(F, J)| \leq \frac{\|S - S_0\|_\infty}{N} \|F\|_{\ell^2(\mathbb{Z})} \|J\|_{\ell^2(\mathbb{Z})} \ll H^{-1/3}, \quad (6.24)$$

because each of the last two norms is at most $\sqrt{N}$.

Every term in $\mathcal{B}_k(F, J)$ contains $F(x)J(x+m^2)$ with $m \geq 1$, and this product is zero by hypothesis. Thus $\mathcal{B}_k(F, J) = 0$ term by term, regardless of the signed weights in (6.7). For a pair x, y with $F(x)J(y) \neq 0$, the ordering assumption gives $1 \leq y - x \leq N - 1$, and the residue conditions give $y - x \equiv r \pmod{D}$. Consequently both indicators in (6.6) are automatic for every contributing pair. Its expansion is exactly

$$\begin{aligned} \mathcal{B}_{k_0}(F, J) &= \frac{D}{N^2} \sum_{x, y \in [N]} F(x) J(y) \sum_{\xi \in \Xi} \mathfrak{g}(\xi) e(-\xi(y-x)) \\ &= D \sum_{\xi \in \Xi} \mathfrak{g}(\xi) \hat{F}(-\xi) \hat{J}(\xi). \end{aligned}$$

Combining this with (6.24) proves (6.3). □

Remark 6.2. The normalization for later interval restrictions follows directly from the proposition. Suppose $I_1, I_2 \subseteq [N]$ are nonempty intervals with $\max I_1 < \min I_2$, and let Ξ be the frequency set in (6.3). Suppose F_i is the indicator of the restriction of a square-difference-free set to I_i and a class a_i modulo D , where $a_2 - a_1$ is a strict square class. The coefficients

$$c_i(\xi) = \frac{D}{|I_i|} \sum_{n \in I_i} F_i(n) e(-n\xi) = \frac{DN}{|I_i|} \widehat{F}_i(\xi)$$

satisfy

$$\left| \sum_{\xi \in \Xi} \mathfrak{g}(\xi) c_1(-\xi) c_2(\xi) \right| \ll H^{-1/3} \frac{DN^2}{|I_1||I_2|}.$$

This conclusion uses the integer supports of the restrictions; it requires no pointwise positivity of a Fourier lift.

7 Comparison along square-step progressions

We next compare the product-space functional with the same functional on shorter intervals. Fixing a few prime coordinates makes its low Fourier coefficients exact averages on a residue class. Splitting that class into progressions with square step preserves the forbidden integer differences and permits a change of scale. The comparison proved here uses no induction hypothesis. Restriction to square-step progressions is already a key feature of quantitative square-difference arguments; see the Main Lemma of [14] and [9, Lemma 6.2]. Here it is used to compare the signed multilinear functional itself.

Keep the constants $\gamma = 1/64$ and $\beta = 10^{-5}$ fixed as above. Enlarge the fixed prime threshold P , if necessary, so that the tuple measures are available for every prime $p \geq P$ and

$$\prod_{\substack{p \geq P \\ p \text{ prime}}} (1 + p^{-3}) - 1 \leq \frac{1}{4}, \quad |\mathfrak{g}(\xi)| \leq q(\xi)^{-1/3} \quad (7.1)$$

whenever $q(\xi)$ is squarefree and all its prime factors are at least P . Such a choice is possible: the product converges, while the prime Gauss estimate used in Proposition 6.1 gives

$$\left| \frac{1}{p-1} \sum_{x \in \mathbb{F}_p^\times} e(ax^2/p) \right| \leq \frac{\sqrt{p}+1}{p-1} = \frac{1}{\sqrt{p}-1} \leq p^{-1/3}$$

for all sufficiently large p and all $a \neq 0$ in $\mathbb{F}_p$. Chinese remaindering then gives the second inequality in (7.1), including $q(\xi) = 1$. Define the fixed integer

$$M_0 = 8 \prod_{\substack{3 \leq p < P \\ p \text{ prime}}} p. \quad (7.2)$$

In particular $M_0 \geq 8$.

For every positive integer N , put

$$\begin{aligned} Q &= N^\beta, & H &= N^{\beta/2}, & \mathcal{P}_N &= \{p \text{ prime} : P \leq p \leq Q\}, \\ \tau &= \frac{\beta\gamma}{1000(d+1)}, & \sigma_0 &= \frac{\gamma\tau}{4}. \end{aligned} \quad (7.3)$$

For a square-difference-free set $A \subseteq [N]$, the quantity that will eventually be estimated by induction is

$$Y(N, A) = Z_{\mathcal{P}_N}(\mathcal{L}_{[N], \mathcal{P}_N, Q} 1_A). \quad (7.4)$$

It is nonnegative by Proposition 4.1. We also set

$$f_s(n) = M_0 1_{\{n \equiv s \pmod{M_0}\}} 1_A(n) \quad (s \in \mathbb{Z}/M_0\mathbb{Z}), \quad 1_A = \mathbb{E}_{s \bmod M_0} f_s. \quad (7.5)$$

The factor M_0 in this definition will be retained in the estimates. All constants below may depend on P, d, β, γ and the earlier fixed construction parameters.

Lemma 7.1 (Exact coefficients on a fiber). *Let $N \geq 1$, let $A \subseteq [N]$, and define f_s by (7.5). Let $I \subseteq [N]$ be a nonempty interval, let $B \subseteq \mathcal{P}_N$, and put*

$$q_B = \prod_{p \in B} p, \quad \mathcal{S} = \mathcal{P}_N \setminus B, \quad D = M_0 q_B.$$

Fix $s \bmod M_0$ and $z_B \in X_B$, and let $a \bmod D$ be their combined residue class. If

$$g_I = (\mathcal{L}_{I, \mathcal{P}_N, Q} f_s)(z_B, \cdot),$$

then every character ξ on $X_{\mathcal{S}}$ satisfying $q(\xi)q_B \leq Q$ has coefficient

$$\hat{g}_I(\xi) = \frac{D}{|I|} \sum_{\substack{n \in I \\ n \equiv a \pmod{D}}} 1_A(n) e(-n\xi). \quad (7.6)$$

Here $q_{\emptyset} = 1$, and the hat denotes the normalized Fourier coefficient on the remaining product space.

Proof. Write $\chi_{\xi}(z) = e(z\xi)$ for a product-space character. Each original character has a unique decomposition $\xi + \eta$, where ξ is supported on $\mathcal{S}$ and η on B . Their denominators are coprime, so $q(\xi + \eta) = q(\xi)q(\eta)$. The hypothesis ensures that every character η on X_B is retained in the lift. Orthogonality over this entire character group gives

$$\begin{aligned} \hat{g}_I(\xi) &= \sum_{\eta \text{ on } X_B} \frac{1}{|I|} \sum_{n \in I} f_s(n) e(-n(\xi + \eta)) \chi_{\eta}(z_B) \\ &= \frac{q_B}{|I|} \sum_{\substack{n \in I \\ n \equiv z_p \pmod{p} \ (p \in B)}} f_s(n) e(-n\xi). \end{aligned}$$

Substituting (7.5) proves (7.6). $\square$

Proposition 7.2 (Comparison with a shorter interval). *There are constants $C_{\text{tr}} < \infty$ and N_{tr} with the following property. Let $N \geq N_{\text{tr}}$, let $A \subseteq [N]$ be square-difference-free, and let $B \subseteq \mathcal{P}_N$ satisfy $q_B \leq N^{\tau}$. Put*

$$\mathcal{S} = \mathcal{P}_N \setminus B, \quad D = M_0 q_B, \quad N' = \lceil N/D^2 \rceil, \quad \lambda = \frac{D^2 N'}{N}.$$

For any $s \bmod M_0$ and $z_B \in X_B$, define

$$g = (\mathcal{L}_{[N], \mathcal{P}_N, Q} f_s)(z_B, \cdot).$$

Then $1 \leq N' < N$, and

$$Z_{\mathcal{S}}(g) \leq \lambda^d \max_{\substack{A' \subseteq [N'] \\ A' \text{ square-difference-free}}} Y(N', A') + C_{\text{tr}} N^{-\sigma_0}. \quad (7.7)$$

Moreover, for arbitrary choices $s_v \bmod M_0$ and $z_{B,v} \in X_B$, one for every $v \in V$, the functions

$$g_v = (\mathcal{L}_{[N], \mathcal{P}_N, Q} f_{s_v})(z_{B,v}, \cdot)$$

satisfy

$$|\mathcal{Z}_{\mathcal{S}}((g_v)_{v \in V})| \leq \lambda^d \max_{\substack{A' \subseteq [N'] \\ A' \text{ square-difference-free}}} Y(N', A') + C_{\text{tr}} N^{-\sigma_0}. \quad (7.8)$$

All constants are uniform in N, A, B, s, z_B and the mixed specifications. Finally,

$$1 \leq \lambda < 1 + \frac{D^2}{N}, \quad \lambda^d = 1 + O_d(D^2/N) = 1 + O_{d, M_0}(N^{-1+2\tau}). \quad (7.9)$$

Proof. We first record the parameter inequalities that make both lifts available. Since $\tau < \beta/2$,

$$Hq_B \leq N^{\beta/2+\tau} \leq Q.$$

Also

$$N' \geq \frac{N}{D^2} \geq M_0^{-2} N^{1-2\tau}. \quad (7.10)$$

Because $1/2 - 2\tau > 0$, this lower bound is at least $N^{1/2}$ for sufficiently large N . As $D \geq 8$, we have $N' < N$ for every integer $N \geq 2$. Hence, with $Q' = (N')^\beta$,

$$H \leq Q' \leq Q, \quad N \geq 10Q^6, \quad N' \geq 10(Q')^6 \quad (7.11)$$

once N is sufficiently large. The length conditions follow from $6\beta < 1$ and the uniform growth of N' in (7.10).

Removing large remaining conductors. Split $g = g_{\leq H} + g_{> H}$ according to the denominator on the remaining prime coordinates $\mathcal{S}$. Before specialization, these pieces retain exactly the denominator families

$$\{U \subseteq \mathcal{P}_N : q_U \leq Q, q_{U \setminus B} \leq H\}, \quad \{U \subseteq \mathcal{P}_N : q_U \leq Q, q_{U \setminus B} > H\}.$$

Thus Lemma 5.2 applies to both pieces. For a function u on $X_B \times X_{\mathcal{S}}$, uniform counting gives the specialization bound

$$\|u(z_B, \cdot)\|_r \leq q_B^{1/r} \|u\|_r \quad (r \geq 1);$$

this is also the estimate in Lemma 5.3. Since $|f_s| \leq M_0$, it follows that, for $g_* \in \{g, g_{\leq H}, g_{> H}\}$ and every fixed even r ,

$$\|g_*\|_r \leq M_0 q_B^{1/r} N^{o(1)}. \quad (7.12)$$

All subpower bounds in this proof are uniform over the indicated sets and fibers.

Telescope the d inputs in the diagonal integral from g to $g_{\leq H}$. Each difference has one input $g_{> H}$. Lemma 4.3 and (7.12) imply, for every fixed $\varepsilon > 0$,

$$|Z_{\mathcal{S}}(g) - Z_{\mathcal{S}}(g_{\leq H})| \leq C_{\varepsilon} H^{-\gamma} q_B N^{\varepsilon}. \quad (7.13)$$

Indeed the specialization powers in that lemma add to

$$\frac{1}{2} + \frac{d-1}{2(d-1)} = 1.$$

The factors d and M_0^d are fixed and are included in C_ε . Applying the moment lemma with sufficiently small individual exponents makes the aggregate moment loss at most N^ε .

Splitting the exact fiber into square-step progressions. Let $a \bmod D$ combine s and z_B . By Lemma 7.1, for every remaining ξ with $q(\xi) \leq H$,

$$\widehat{g}(\xi) = \frac{D}{N} \sum_{\substack{n \leq N \\ n \equiv a \pmod{D}}} 1_A(n) e(-n\xi). \quad (7.14)$$

Let

$$J_a = \{j \in [D^2] : j \equiv a \pmod{D}\}, \quad c_j = j - D^2, \\ A'_j = \{n' \in [N'] : D^2 n' + c_j \in A\}.$$

There are exactly D elements of J_a . The map $(n', j) \mapsto D^2(n' - 1) + j$ bijects $[N'] \times [D^2]$ with $[D^2 N']$. Restricting j to J_a selects exactly the residue class $a \bmod D$. The definition of A'_j assigns zero to all padded points greater than N , so this decomposition incurs no additive error.

Each A'_j is square-difference-free: a difference $m^2 > 0$ between its elements would give the difference $D^2 m^2 = (Dm)^2$ between the corresponding elements of A . This argument concerns actual integer squares.

For $p \in \mathcal{S}$, the integer D is invertible modulo p . Define

$$T_j(z)_p = D^{-2}(z_p - c_j) \pmod{p}, \quad h_j^H = \mathcal{L}_{[N'], \mathcal{S}, H} 1_{A'_j}.$$

Each T_j is a translation followed by multiplication by the nonzero square $(D^{-1})^2$. Multiplication of a remaining frequency by D^2 preserves its exact denominator, so changing variables $\eta = D^2 \xi$ in the Fourier expansion of (7.14) gives the exact identity

$$g_{\leq H}(z) = \frac{\lambda}{D} \sum_{j \in J_a} h_j^H(T_j(z)). \quad (7.15)$$

For clarity, the coefficient of $e(z\xi)$ on the right is

$$\frac{DN'}{N} \sum_{j \in J_a} e(-c_j \xi) \left(\frac{1}{N'} \sum_{n' \leq N'} 1_{A'_j}(n') e(-D^2 n' \xi) \right),$$

which is exactly (7.14).

The seminorm triangle inequality and affine invariance in Lemma 4.2 now imply

$$Z_{\mathcal{S}}(g_{\leq H})^{1/d} \leq \frac{\lambda}{D} \sum_{j \in J_a} Z_{\mathcal{S}}(h_j^H)^{1/d} \leq \lambda \max_{j \in J_a} Z_{\mathcal{S}}(h_j^H)^{1/d}.$$

Consequently,

$$Z_{\mathcal{S}}(g_{\leq H}) \leq \lambda^d \max_{j \in J_a} Z_{\mathcal{S}}(h_j^H). \quad (7.16)$$

The averaging by D is essential: the number of progressions causes no further loss.

Recovering the smaller-scale functional. Set

$$h_j^{Q'} = \mathcal{L}_{[N'], \mathcal{S}, Q'} 1_{A'_j}.$$

The difference $h_j^{Q'} - h_j^H$ has conductor greater than H . By (7.11), the moment lemma applies at scale N' to these lifts and their pieces. A second telescoping application of Lemma 4.3, this time without specialization, gives

$$|Z_S(h_j^H) - Z_S(h_j^{Q'})| \leq C_\varepsilon H^{-\gamma} (N')^\varepsilon \quad (7.17)$$

uniformly in j .

The lift $h_j^{Q'}$ depends only on coordinates in $\mathcal{S}' = \mathcal{P}_{N'} \setminus B$, because every prime in a denominator at most Q' is itself at most Q' . On the other hand, the full lift

$$u'_j = \mathcal{L}_{[N'], \mathcal{P}_{N'}, Q'} 1_{A'_j}$$

becomes exactly $h_j^{Q'}$ after uniform averaging over $B \cap \mathcal{P}_{N'}$. Character orthogonality removes precisely the denominators containing one of those primes. By Lemma 4.2, coordinate averaging contracts the seminorm; unused probability factors integrate to one. Therefore

$$Z_S(h_j^{Q'}) = Z_{\mathcal{S}'}(h_j^{Q'}) \leq Z_{\mathcal{P}_{N'}}(u'_j) = Y(N', A'_j). \quad (7.18)$$

This also covers primes of B larger than Q' , which are absent from $\mathcal{P}_{N'}$ already.

Combining (7.13), (7.16), (7.17), and (7.18) proves (7.7) with error

$$C_\varepsilon \left(H^{-\gamma} q_B N^\varepsilon + \lambda^d H^{-\gamma} (N')^\varepsilon \right). \quad (7.19)$$

Uniform error bounds and mixed inputs. The definition of N' gives $1 \leq \lambda < 1 + D^2/N$, and

$$D^2/N \leq M_0^2 N^{-1+2\tau} = o(1).$$

Since d is fixed, these inequalities prove (7.9) and show that λ^d is bounded for large N . Choose $\varepsilon = \beta\gamma/8$ in (7.19). Its two terms, before the bounded factor λ^d , are at most

$$N^{-3\beta\gamma/8+\tau} \quad \text{and} \quad N^{-3\beta\gamma/8},$$

respectively. As $\tau \leq \beta\gamma/1000$ and $\sigma_0 = \gamma\tau/4 \leq \tau/4$, they are $O(N^{-\sigma_0})$. This establishes the diagonal bound with constants independent of any inductive estimate for Y .

Every g_v in the mixed statement is real and satisfies the same diagonal bound. Its right-hand side is nonnegative and independent of v . Proposition 4.1 therefore gives (7.8). $\square$

In particular, the mixed estimate holds pointwise after specifying all labels at primes in B , regardless of the law used to choose those labels. It can therefore be integrated against any such probability law. When the prime laws form a product, the remaining coordinates still carry the full laws used in Proposition 7.2. No reflection positivity is required of the law on the specified labels.

8 Contraction on ordered intervals

A tuple of residue classes $\mathbf{s} = (s_v)_{v \in V} \in (\mathbb{Z}/M_0\mathbb{Z})^V$ is called *good* if

$$s_{v_{j+1}} - s_{v_j} \quad \text{is a strict square class modulo } M_0 \quad (0 \leq j < 24),$$

where $v_{24} = v_0$. Thus each difference is 1 modulo 8 and a nonzero square at every odd prime dividing M_0 . Let

$$\rho = M_0^{-d} |\{\mathbf{s} \in (\mathbb{Z}/M_0\mathbb{Z})^V : \mathbf{s} \text{ is good}\}|. \quad (8.1)$$

We will bound the contribution of each good tuple when every prime uses the nonexceptional part of its measure. First we show that these assignments have a fixed positive proportion; the proof also explains the cycle length 24. We retain the parameters and lifts from the preceding section.

Lemma 8.1 (Existence of good residue assignments). *The proportion ρ defined in (8.1) satisfies*

$$\rho \geq M_0^{-23} > 0.$$

Proof. We construct a length-24 sequence of admissible increments with sum zero at each factor of M_0 . Modulo 8, take all the increments equal to 1; their sum is $24 \equiv 0 \pmod{8}$.

Let $p < P$ be an odd prime. If -1 is a square modulo p , repeat the pair $(1, -1)$ twelve times. Both entries are nonzero squares and each pair has sum zero. If -1 is not a square, put

$$S = \{x^2 : x \in \mathbb{F}_p\}.$$

The sets S and $-1 - S$ each have $(p+1)/2$ elements, so they intersect. Thus $-1 = a + b$ for some $a, b \in S$. Neither a nor b is zero, since otherwise -1 would be a square. The triple $(a, b, 1)$ consists of nonzero squares and has sum zero. Repeat this triple eight times. This also covers $p = 3$, where the triple is $(1, 1, 1)$.

The Chinese remainder theorem now gives increments $\Delta_0, \dots, \Delta_{23} \pmod{M_0}$, each a strict square class, with $\sum_{j=0}^{23} \Delta_j = 0$. For each initial residue $a \pmod{M_0}$, prescribe

$$s_{v_0} = a, \quad s_{v_j} = a + \sum_{i=0}^{j-1} \Delta_i \quad (1 \leq j < 24).$$

The last edge has increment Δ_{23} because the sum of all the increments is zero. The slots $v_0, \dots, v_{23}$ are distinct, since their block words are the distinct cyclic shifts of the identity word. Repetition of residue values at nonadjacent slots causes no difficulty. The other $d-24$ residues are unrestricted. We have therefore exhibited at least $M_0 \cdot M_0^{d-24}$ good assignments among the M_0^d possible assignments, which proves the bound. $\square$

The length 24 is divisible by 8, 2, and 3, exactly the three local cycle lengths used in this construction. Thus this choice works uniformly for every prime included in the fixed modulus.

We now prove the analytic estimate for each good assignment.

Proposition 8.2 (Contraction for good residue assignments). *There are absolute constants C and N_0 such that, whenever $N \geq N_0$, $A \subseteq [N]$ has no positive square difference, and $\mathbf{s}$ is good, one has*

$$\left| \int \prod_{v \in V} \mathcal{L}_{[N], \mathcal{P}_N, Q} f_{s_v}(z_v) d \bigotimes_{p \in \mathcal{P}_N} (\mu_p - \varepsilon_p) \right| \leq C N^{-\sigma_0}. \quad (8.2)$$

The constants are uniform in A and $\mathbf{s}$.

Proof. Write η_p for the total mass of ε_p , and set

$$\mu_p^\circ = \frac{\mu_p - \varepsilon_p}{1 - \eta_p}, \quad \Lambda^\circ = \bigotimes_{p \in \mathcal{P}_N} \mu_p^\circ.$$

It suffices to prove the assertion with Λ° in place of the measure in (8.2), since the latter is $\prod_p (1 - \eta_p) \Lambda^\circ$ and $0 \leq \prod_p (1 - \eta_p) \leq 1$. The single-slot marginals of Λ° are uniform. Moreover, at each prime its marginal on any directed cycle edge is uniform on strict square pairs. We use conductor decay for Λ° , without any claim that this normalized measure is reflection positive.

Selecting an ordered pair. Put $K = \lfloor N^\tau \rfloor$, and partition $[N]$ into ordered consecutive intervals $I_1, \dots, I_K$ of sizes differing by at most one. Writing $w_i = |I_i|/N$, we have the exact identity

$$\mathcal{L}_{[N], \mathcal{P}_N, Q} f_s = \sum_{i=1}^K w_i \mathcal{L}_{I_i, \mathcal{P}_N, Q} f_s.$$

Thus the integral is an average over independent interval choices in all slots, with probabilities w_i . For large N , every interval has length at least $\frac{1}{2}N^{1-\tau}$. Since $1 - \tau > 6\beta$, this length is at least $10Q^6$. Lemma 5.2 therefore applies uniformly to every interval lift and every exact-denominator truncation used below. Its application to f_s , which is bounded by the fixed constant M_0 , changes only the constants. All occurrences of $N^{o(1)}$ in this proof are uniform over these interval choices, residues and sets.

First consider assignments for which all 24 cycle slots choose the same interval. Their total probability is

$$\sum_{i=1}^K w_i^{24} \leq (\max_i w_i)^{23} \ll N^{-23\tau}.$$

For every assignment, ordinary Hölder and the uniform single-slot marginals give

$$\left| \int \prod_v G_v(z_v) d\Lambda^\circ \right| \leq \prod_v \|G_v\|_d = N^{o(1)}, \quad G_v = \mathcal{L}_{I(v), \mathcal{P}_N, Q} f_{s_v}.$$

Here d is even, so the required moment is supplied directly by Lemma 5.2. The contribution of these assignments is therefore $O(N^{-23\tau+o(1)})$; there is no restriction on the interval choices in the other slots.

In every remaining assignment there is a directed cycle edge $u \rightarrow v$ whose first interval precedes its second. Indeed, if the cyclic sequence of interval indices had no ascent, it would be nonincreasing all the way around the cycle, and hence constant. Fix one ascending edge, and denote its two intervals by I_1, I_2 for the rest of this argument. All integers in I_1 precede all integers in I_2 .

Eliminating the other slots. Retain these two lifts in full. In each of the other $d - 2$ slots, truncate the lift to exact conductors at most $T = N^\tau$. Lemma 4.3, applied during a telescoping replacement of these slots, bounds the total error by

$$O(T^{-\gamma} N^{o(1)}) = O(N^{-\gamma\tau+o(1)}). \quad (8.3)$$

Every norm used here is controlled by Lemma 5.2: the retained and discarded pieces are subfamilies of complete exact denominator pieces. This error is estimated before the Fourier expansion that follows.

Each Fourier coefficient in a truncated slot has modulus at most M_0 , and there are at most $\sum_{q \leq T} \varphi(q) \ll T^2$ frequencies. The product expansion of all other slots consequently has total coefficient absolute sum at most

$$C_{d, M_0} T^{2(d-2)} \ll N^{2d\tau}. \quad (8.4)$$

For one term in this expansion, let $B \subseteq \mathcal{P}_N$ be the union of the prime supports of its characters, and put

$$R = \prod_{p \in B} p.$$

The conductors are squarefree, so their least common multiple satisfies

$$R \leq T^{d-2} \leq N^{d\tau}. \quad (8.5)$$

Condition on all slot labels at the primes in B . The expanded characters become constants of modulus one. Since Λ° is a product over primes, the tuple laws at the remaining primes are unchanged and independent. The two selected labels now have fixed coordinates $a_{1,B}, a_{2,B}$ at B , whose difference is a nonzero square at each prime in B . At every remaining prime their pair marginal is still uniform on strict square pairs.

Let $\mathcal{S} = \mathcal{P}_N \setminus B$, let $s_1 = s_u$ and $s_2 = s_v$, and define functions on $X_{\mathcal{S}}$ by

$$g_i(z) = (\mathcal{L}_{I_i, \mathcal{P}_N, Q} f_{s_i})(a_{i,B}, z), \quad i = 1, 2.$$

With Fourier coefficients normalized for uniform measure on $X_{\mathcal{S}}$, their conditional pair integral is exactly

$$\sum_{\xi \text{ on } X_{\mathcal{S}}} \widehat{g}_1(-\xi) \widehat{g}_2(\xi) \mathfrak{g}(\xi). \quad (8.6)$$

To see this, simultaneous translation kills character pairs whose frequencies do not sum to zero. For a matching pair, the multiplier is the character average of the difference of the labels. At every odd prime a uniform nonzero square is the square of a uniform nonzero unit. The Chinese remainder theorem therefore identifies this average with the unit-square multiplier $\mathfrak{g}(\xi)$ used in Proposition 6.1.

Estimating the selected pair. Specialization on a uniform coordinate block of size R costs at most $R^{1/2}$ in L^2 : for every function G ,

$$\|G(a_B, \cdot)\|_2^2 \leq R \|G\|_2^2.$$

This inequality holds at every specified fiber, independently of the law under which that fiber was selected. Hence $\|g_i\|_2 \leq R^{1/2} N^{o(1)}$. By the bound $|\mathfrak{g}(\xi)| \leq q(\xi)^{-1/3}$, Parseval and Cauchy-Schwarz, the part of (8.6) with conductor exceeding H has modulus at most

$$H^{-1/3} \|g_1\|_2 \|g_2\|_2 \leq H^{-1/3} R N^{o(1)}. \quad (8.7)$$

We next identify the remaining coefficients with actual integer fibers. Put $D = M_0 R$. The parameter inequalities

$$d\tau < \frac{\beta\gamma}{1000} < \frac{\beta}{2} \quad (8.8)$$

imply $RH \leq Q$ and, for sufficiently large N , $D \leq N^\beta$. For a remaining frequency ξ with $q(\xi) \leq H$, all frequencies $\xi + \eta$, where η ranges over the character group on B , therefore occur in the original lift. Summing this entire character group gives

$$\begin{aligned} \widehat{g}_i(\xi) &= \sum_{\eta \text{ on } B} \frac{1}{|I_i|} \sum_{n \in I_i} f_{s_i}(n) e(-n(\xi + \eta)) e(a_{i,B}\eta) \\ &= \frac{D}{|I_i|} \sum_{\substack{n \in I_i \\ n \equiv a_i \pmod{D}}} 1_A(n) e(-n\xi), \end{aligned} \quad (8.9)$$

where $a_i \pmod{D}$ combines $s_i \pmod{M_0}$ and the coordinates $a_{i,B}$ by the Chinese remainder theorem. Character orthogonality supplies the factor R , while the definition of f_{s_i} supplies M_0 . Goodness of $\mathbf{s}$ and the constraints at the conditioned primes show that $a_2 - a_1$ is a strict square class modulo D .

The remaining frequencies of conductor at most H are precisely the frequencies with squarefree conductor at most H coprime to D . Indeed, D contains every prime less than P ; every prime divisor of such a conductor is consequently an available prime at least P , at most $H \leq Q$, and outside B . This includes the zero frequency under the convention $q(0) = 1$.

Define functions on $[N]$ by

$$F_i(n) = 1_A(n)1_{I_i}(n)1_{n \equiv a_i \pmod{D}}.$$

They are bounded by one, their supports are ordered, and there is no positive square difference from the first support to the second. Moreover, D is eight times an odd squarefree number and $D \leq N^\beta$. All hypotheses of Proposition 6.1 hold. If the Fourier coefficients of F_i use normalization N^{-1} , then (8.9) gives $\widehat{g}_i(\xi) = DN\widehat{F}_i(\xi)/|I_i|$. Thus the part of (8.6) of conductor at most H is bounded in modulus by

$$\begin{aligned} \frac{D^2 N^2}{|I_1||I_2|} \left| \sum_{\substack{q(\xi) \leq H, \ q(\xi) \text{ squarefree} \\ (q(\xi), D) = 1}} \mathfrak{g}(\xi) \widehat{F}_1(-\xi) \widehat{F}_2(\xi) \right| \\ \ll H^{-1/3} D \frac{N^2}{|I_1||I_2|} \ll H^{-1/3} D N^{2\tau}. \end{aligned} \quad (8.10)$$

The factor D in Proposition 6.1 has canceled one of the two fiber-normalization factors.

Combining (8.7) and (8.10) bounds each conditional pair integral by $O(N^{-\beta/6+(d+2)\tau+o(1)})$. This is uniform over all conditionings of positive probability. Averaging them, and then summing the Fourier expansion using (8.4), bounds the truncated integral for each unequal-cycle interval assignment by

$$O(N^{-\beta/6+(3d+2)\tau+o(1)}).$$

Averaging interval assignments incurs no further loss. Including the equal-cycle contribution and (8.3), the original integral has modulus at most

$$C \left(N^{-23\tau+o(1)} + N^{-\gamma\tau+o(1)} + N^{-\beta/6+(3d+2)\tau+o(1)} \right). \quad (8.11)$$

Finally, $23\tau = 5888\sigma_0$, $\gamma\tau = 4\sigma_0$, and

$$\frac{\beta}{6} - (3d+2)\tau > \beta \left(\frac{1}{6} - \frac{3}{64000} \right) > \frac{\beta}{7} > 2\sigma_0.$$

The finitely many uniform moment losses in each term can be bounded by N^{σ_0} , after increasing the absolute threshold in N . Equation (8.11) is consequently $O(N^{-\sigma_0})$, as required. All estimates used absolute values of signed Fourier lifts; no pointwise positivity of a lift was needed. $\square$

9 Exceptional terms and the power-saving induction

We now combine the scale comparison, which applies to every small-residue assignment, with the cancellation obtained for good assignments. For good assignments the nonexceptional contribution is already negligible; we control the exceptional parts by expanding the prime measures. The other assignments are bounded directly by the scale comparison. Expanding measures rather than comparing integrands pointwise keeps the argument valid for signed Fourier lifts throughout.

The finite set V and the constants $P, M_0, \beta, \gamma, \tau, \sigma_0$ have now been fixed. Lemma 8.1 has also established the fixed positive proportion ρ of good small-residue assignments. The prime threshold P was chosen so that

$$\prod_{p \geq P} (1 + p^{-3}) - 1 \leq \frac{1}{4}. \quad (9.1)$$

For every positive integer n , define

$$\mathcal{Y}(n) = \max_{\substack{A \subseteq [n] \\ A \text{ square-difference-free}}} Y(n, A). \quad (9.2)$$

This is a finite nonnegative number, by Proposition 4.1. Write $\eta_p = \varepsilon_p(\mathbb{F}_p^V)$ for the mass of the exceptional measure, so $0 \leq \eta_p \leq p^{-4}$. For $B \subseteq \mathcal{P}_N$, put

$$q_B = \prod_{p \in B} p, \quad \eta_B = \prod_{p \in B} \eta_p, \quad n_B = \left\lceil \frac{N}{(M_0 q_B)^2} \right\rceil. \quad (9.3)$$

Empty products have value 1, so $q_\emptyset = \eta_\emptyset = 1$ and $n_\emptyset = \lceil N/M_0^2 \rceil$.

Proposition 9.1. *There are absolute constants $K < \infty$ and $N_1 \geq 2$ such that, for every integer $N \geq N_1$,*

$$\mathcal{Y}(N) \leq K N^{-\sigma_0} + (1 + u_N) \left((1 - \rho) \mathcal{Y}(n_\emptyset) + \rho \sum_{\substack{\emptyset \neq B \subseteq \mathcal{P}_N \\ q_B \leq N^\tau}} \eta_B \mathcal{Y}(n_B) \right), \quad (9.4)$$

$$u_N = (1 + M_0^2 N^{2\tau-1})^d - 1. \quad (9.5)$$

In particular, $u_N \rightarrow 0$ as $N \rightarrow \infty$. The constants K, N_1 do not depend on any proposed induction bound for $\mathcal{Y}$.

Proof. Fix a square-difference-free set $A \subseteq [N]$. Write

$$G_s = \mathcal{L}_{[N], \mathcal{P}_N, Q} f_s, \quad \Phi_{\mathbf{s}}((z_v)_{v \in V}) = \prod_{v \in V} G_{s_v}(z_v),$$

where $\mathbf{s} = (s_v)_{v \in V}$ is a small-residue assignment. By (7.5), $1_A = \mathbb{E}_s f_s$. The lift is linear, and the product-space integral is multilinear, so

$$Y(N, A) = \mathbb{E}_{\mathbf{s}} \int \Phi_{\mathbf{s}} d \bigotimes_{p \in \mathcal{P}_N} \mu_p. \quad (9.6)$$

For $B \subseteq \mathcal{P}_N$, define the nonnegative finite measure

$$\kappa_B = \bigotimes_{p \in B} \varepsilon_p \otimes \bigotimes_{p \in \mathcal{P}_N \setminus B} \mu_p.$$

Its mass is η_B . The finite product expansion gives the following identity of signed measures:

$$\bigotimes_{p \in \mathcal{P}_N} \mu_p - \bigotimes_{p \in \mathcal{P}_N} (\mu_p - \varepsilon_p) = \sum_{\emptyset \neq B \subseteq \mathcal{P}_N} (-1)^{|B|+1} \kappa_B. \quad (9.7)$$

For any real integrand Φ , integration followed by the triangle inequality therefore yields

$$\int \Phi d \bigotimes_p \mu_p \leq \left| \int \Phi d \bigotimes_p (\mu_p - \varepsilon_p) \right| + \sum_{B \neq \emptyset} \left| \int \Phi d \kappa_B \right|. \quad (9.8)$$

This step makes no sign assumption on Φ . Notice also that the measure outside B in each summand is the full measure μ_p .

We first bound the summands with $q_B > N^\tau$. A summand of zero mass is zero, so assume $\eta_B > 0$. Each normalized exceptional measure ε_p/η_p has uniform single-label marginals: its invariance under simultaneous translation implies that any one label is uniformly distributed on $\mathbb{F}_p$. The same uniformity holds for μ_p . Consequently every single-slot marginal of κ_B/η_B is uniform on $X_{\mathcal{P}_N}$. No independence between different slots is needed. Ordinary Hölder's inequality gives

$$\left| \int \Phi_{\mathbf{s}} d\kappa_B \right| \leq \eta_B \prod_{v \in V} \|G_{s_v}\|_{L^d(X_{\mathcal{P}_N})} \leq C_\epsilon \eta_B N^\epsilon \quad (\epsilon > 0). \quad (9.9)$$

The last inequality is Lemma 5.2, applied at the fixed even order d and with the loss in each factor chosen sufficiently small. The bound $|f_s| \leq M_0$ only introduces a fixed factor. The constant C_ϵ is independent of B , A , N , and $\mathbf{s}$.

The exceptional masses are summable with the extra factor q_B :

$$\sum_{\emptyset \neq B \subseteq \mathcal{P}_N} q_B \eta_B = \prod_{p \in \mathcal{P}_N} (1 + p\eta_p) - 1 \leq \prod_{p \geq P} (1 + p^{-3}) - 1 \leq \frac{1}{4}. \quad (9.10)$$

It follows from (9.9) that

$$\sum_{\substack{\emptyset \neq B \subseteq \mathcal{P}_N \\ q_B > N^\tau}} \left| \int \Phi_{\mathbf{s}} d\kappa_B \right| \leq \frac{C_\epsilon}{4} N^{-\tau+\epsilon}.$$

Choose $\epsilon = \tau/2$. Since $\sigma_0 = \gamma\tau/4 < \tau/2$, we obtain

$$\sum_{q_B > N^\tau} \left| \int \Phi_{\mathbf{s}} d\kappa_B \right| \leq K_1 N^{-\sigma_0}, \quad (9.11)$$

uniformly in the residue assignment. Thus the number of exceptional subsets has already been accounted for by the convergent product (9.10).

Now suppose $q_B \leq N^\tau$ and $\eta_B > 0$. Condition on all the labels at primes in B under κ_B/η_B . The remaining prime law is exactly $\bigotimes_{p \in \mathcal{P}_N \setminus B} \mu_p$. For each realization, the function in slot v is G_{s_v} specialized at its prescribed labels in B . The mixed assertion of Proposition 7.2 applies to these arbitrary slotwise residues and labels. With

$$D_B = M_0 q_B, \quad \lambda_B = \frac{D_B^2 n_B}{N},$$

it gives

$$\left| \int \Phi_{\mathbf{s}} d\kappa_B \right| \leq \eta_B \left(\lambda_B^d \mathcal{Y}(n_B) + K_2 N^{-\sigma_0} \right). \quad (9.12)$$

The same inequality is trivially true when $\eta_B = 0$. Crucially, the error and its lower threshold for N are uniform in B and in every conditioned tuple. No induction hypothesis has been used.

The ceiling in the smaller scale produces a uniform multiplicative error. For $q_B \leq N^\tau$,

$$1 \leq \lambda_B \leq 1 + \frac{D_B^2}{N} \leq 1 + M_0^2 N^{2\tau-1},$$

and hence $\lambda_B^d \leq 1 + u_N$. Moreover, $\sum_{B \neq \emptyset} \eta_B \leq 1/4$ by (9.10). Summing the errors in (9.12) is therefore harmless.

If $\mathbf{s}$ is good, Proposition 8.2 bounds the first term on the right of (9.8) by $K_3 N^{-\sigma_0}$. Combining this with (9.11) and (9.12), we obtain

$$\int \Phi_{\mathbf{s}} d \bigotimes_p \mu_p \leq K_4 N^{-\sigma_0} + (1 + u_N) \sum_{\substack{\emptyset \neq B \subseteq \mathcal{P}_N \\ q_B \leq N^\tau}} \eta_B \mathcal{Y}(n_B).$$

For a nongood $\mathbf{s}$, apply the mixed scale comparison directly with $B = \emptyset$:

$$\left| \int \Phi_{\mathbf{s}} d \bigotimes_p \mu_p \right| \leq (1 + u_N) \mathcal{Y}(n_{\emptyset}) + K_2 N^{-\sigma_0}.$$

Averaging these two estimates in (9.6) gives (9.4), since the good assignments have probability ρ . The estimate is uniform in A , so we may take its maximum. Finally, $2\tau < 1$ by the fixed parameter choice, and M_0, d are fixed, so $u_N \rightarrow 0$. All constants in the proof came from the preceding estimates and fixed parameters; none depended on an induction constant. $\square$

Theorem 9.2. *There are absolute constants $a_0 > 0$ and $C_0 < \infty$ such that, for every positive integer N and every square-difference-free set $A \subseteq [N]$,*

$$Y(N, A) \leq C_0 N^{-a_0}.$$

Proof. We make all remaining parameter choices before starting the induction. The finite construction fixes h, t_0, ℓ, d ; the conditional and kernel estimates fix γ and β . The threshold P was then chosen, and with it M_0, τ , and σ_0 . In particular the probability ρ of Lemma 8.1 is already a fixed positive number. We shall not enlarge P after choosing the exponent below.

Put $r = 1 - 3\rho/4$, so $1/4 \leq r < 1$, and choose

$$a_0 = \min \left\{ \frac{1}{4}, \frac{\sigma_0}{2}, \frac{\log(1/r)}{4 \log M_0} \right\}. \quad (9.13)$$

Table 1 summarizes the roles of the parameters; among the listed quantities, only those in the last row vary with the interval length.

Parameters	Role and order of choice
h, t_0, ℓ, d	Fixed cycle length, marking depth, block count, and tuple size (Section 3).
γ, β	Fixed conductor-decay and rational-frequency cutoff exponents (Sections 4 and 6).
P, M_0	Prime-law threshold and modulus containing all smaller primes; fixed before choosing the decay exponent (Section 7).
τ, σ_0	Fixed partition exponent and common error exponent; chosen in (7.3).
ρ, a_0	Fixed positive residue proportion and induction exponent; a_0 is chosen only after M_0 and ρ in (9.13).
$Q, H, \mathcal{P}_N$	Scale-dependent cutoffs and prime set, defined from N in (7.3).

Table 1: Parameter roles. Every construction size and exponent is fixed independently of N and A ; the Fourier cutoffs and available prime coordinates change with N .

Every quantity in (9.13) is fixed and positive. Thus $0 < a_0 < \min(1/2, \sigma_0)$, and

$$\theta := M_0^{2a_0} (1 - \rho + \rho/4) = M_0^{2a_0} r \leq \sqrt{r} < 1. \quad (9.14)$$

There is no requirement that a_0 be numerically useful.

For every finite $\mathcal{P}_N$, the bound $\eta_p \leq p^{-4}$ and $2a_0 \leq 1$ give

$$\sum_{\emptyset \neq B \subseteq \mathcal{P}_N} \eta_B q_B^{2a_0} = \prod_{p \in \mathcal{P}_N} (1 + \eta_p p^{2a_0}) - 1 \leq \prod_{p \geq P} (1 + p^{-3}) - 1 \leq \frac{1}{4}. \quad (9.15)$$

Let $\kappa = (1 - \theta)/2 > 0$. Choose an integer $N_* \geq N_1$ so large that, for every $N \geq N_*$,

$$\theta u_N \leq \kappa, \quad KN^{a_0 - \sigma_0} \leq \kappa, \quad (9.16)$$

where K, N_1 are supplied by Proposition 9.1. This is possible because $u_N \rightarrow 0$ and $a_0 < \sigma_0$. The choice of N_* is independent of C_0 . Choose $C_0 \geq 1$ large enough that

$$\mathcal{Y}(n) \leq C_0 n^{-a_0} \quad (1 \leq n < N_*). \quad (9.17)$$

Only finitely many values are involved, so such a finite constant exists. For an elementary bound, the lift in $Y(n, A)$ has at most $2n^{2\beta}$ frequencies and each coefficient has modulus at most 1. Since its tuple law is a probability measure, $0 \leq Y(n, A) \leq (2n^{2\beta})^d$. Thus, for example, $C_0 = \max\{1, 2^d N_*^{2\beta d + a_0}\}$ suffices in (9.17). This also covers $n = 1$ and the cases of an empty prime set.

Suppose now that $N \geq N_*$ and $\mathcal{Y}(n) \leq C_0 n^{-a_0}$ for every positive integer $n < N$. Every smaller scale appearing in (9.4) lies in this range. Indeed, $M_0 q_B \geq 8$ gives

$$1 \leq n_B \leq N/64 + 1 < N \quad (N \geq 2).$$

The ceiling goes in the favorable direction for a negative exponent:

$$n_B^{-a_0} \leq N^{-a_0} M_0^{2a_0} q_B^{2a_0}.$$

Apply the induction hypothesis at every such scale in (9.4), and then use (9.15). Extending the positive subset sum to all nonempty B can only increase the bound, and therefore

$$\begin{aligned} \mathcal{Y}(N) &\leq KN^{-\sigma_0} + (1 + u_N)C_0 N^{-a_0} M_0^{2a_0} \left(1 - \rho + \rho \sum_{B \neq \emptyset} \eta_B q_B^{2a_0}\right) \\ &\leq KN^{-\sigma_0} + (1 + u_N)\theta C_0 N^{-a_0} \\ &\leq \kappa N^{-a_0} + (1 - \kappa)C_0 N^{-a_0} \leq C_0 N^{-a_0}. \end{aligned}$$

In the third line we used (9.16) and $\theta + \kappa = 1 - \kappa$; in the last line we used $C_0 \geq 1$. This closes strong induction. In particular, increasing C_0 to cover the finite base range does not change the validity of the large-scale step. $\square$

Proof of Theorem 1.1. Let $N \geq 1$ and let $A \subseteq [N]$ have no positive integer-square difference. Put $G = \mathcal{L}_{[N], \mathcal{P}_N, \mathbb{Q}} 1_A$. Its constant Fourier coefficient is $|A|/N$, and every nonconstant character on $X_{\mathcal{P}_N}$ has uniform mean zero. The mean inequality in Lemma 4.2 and Theorem 9.2 yield

$$\frac{|A|}{N} = |\mathbb{E}G| \leq Z_{\mathcal{P}_N}(G)^{1/d} = Y(N, A)^{1/d} \leq C_0^{1/d} N^{-a_0/d}.$$

Thus $C = C_0^{1/d} < \infty$ and $c = a_0/d > 0$ are absolute constants for which $|A| \leq CN^{1-c}$ for every such N and A . $\square$

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